Geo-Mamba

Medical Image Analysis 112 (2026) 104081

Contents lists available at ScienceDirect

Medical Image Analysis

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Geo-Mamba: Geometry-informed state-space learning of functional brain organization

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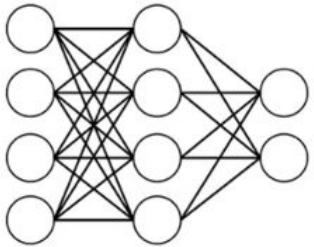
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[b] Departments of Psychiatry and Computer Science, University of North Carolina at Chapel Hill, Chapel Hill, 27599, NC, USA

Outline

1. Architecture Basics
2. Results of Different Architectures
3. Geo-Mamba Architecture
    a. Functional Connectivity Matrices
    b. SPDNet
    c. Geometric Attention Pooling
    d. SSM - Sequence Transduction
4. Ablation Studies
5. Other Analyses

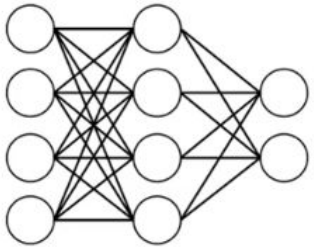
Guiding Question: What inductive biases do our design choices introduce?

Architecture



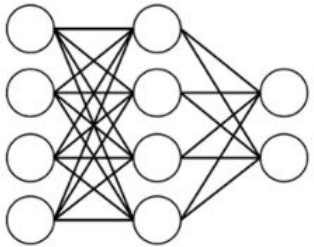
Perceptrons

Architecture



Perceptrons
Function regularity

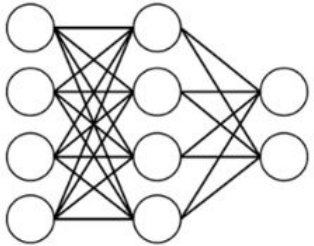
Architecture



Perceptrons
Function regularity

Translation

Architecture

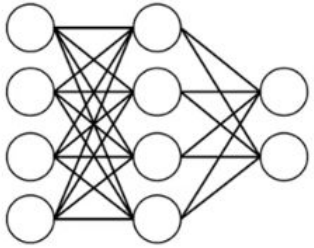


Perceptrons
Function regularity

Translation

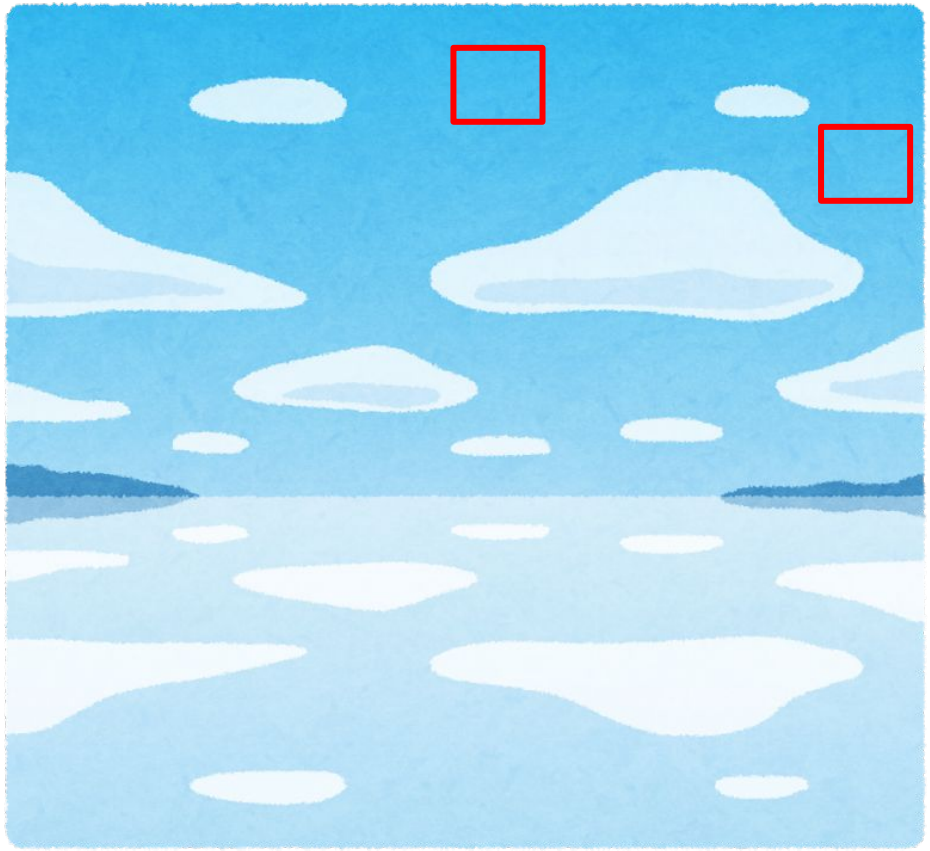


Architecture

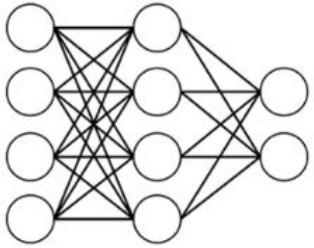


Perceptrons
Function regularity

Translation

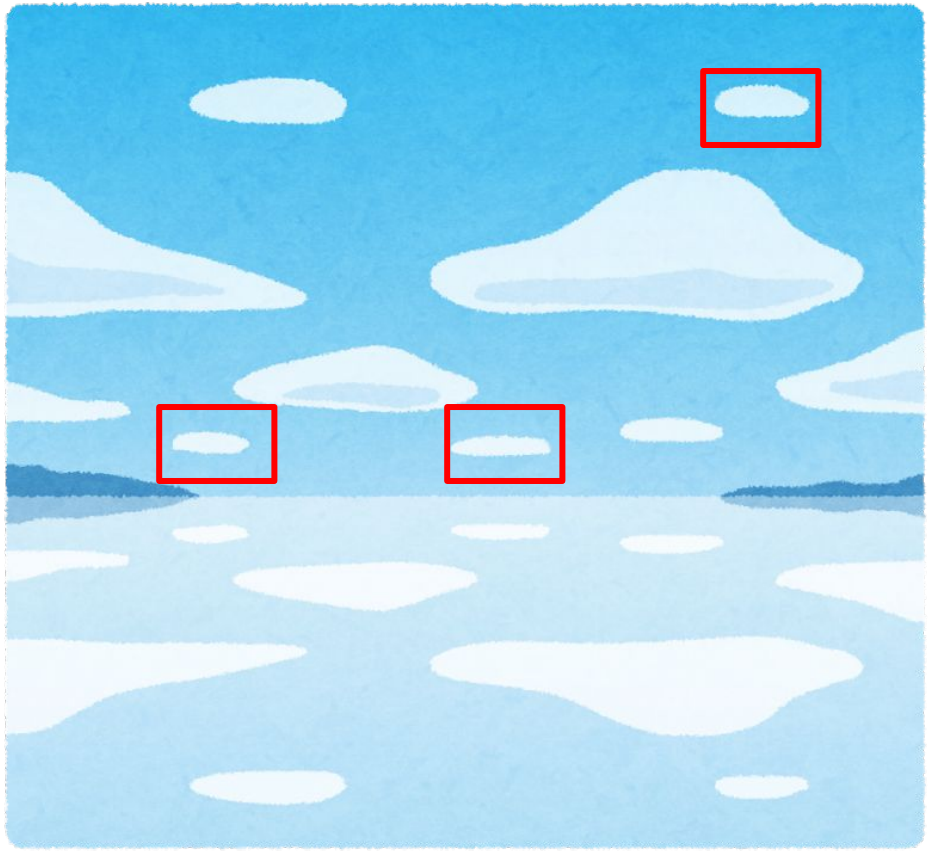


Architecture



Perceptrons
Function regularity

Translation



Architecture

ImageNet Classification with Deep Convolutional Neural Networks

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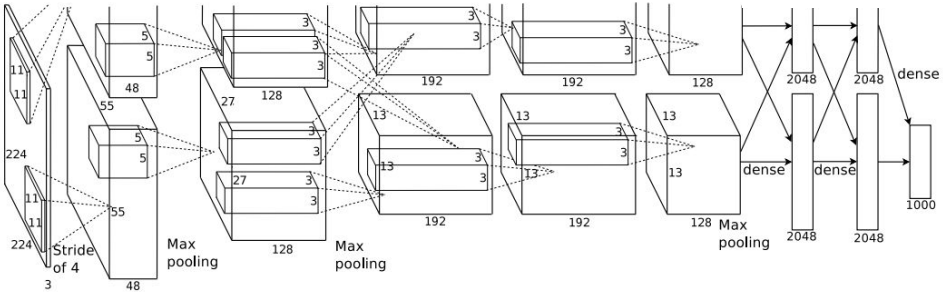
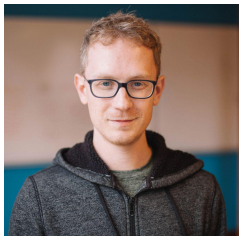


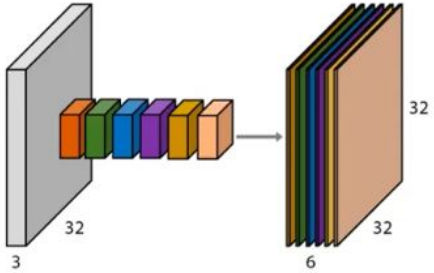
Figure 2: An illustration of the architecture of our CNN, explicitly showing the delineation of responsibilities between the two GPUs. One GPU runs the layer-parts at the top of the figure while the other runs the layer-parts at the bottom. The GPUs communicate only at certain layers. The network's input is 150,528-dimensional, and the number of neurons in the network's remaining layers is given by 253,440–186,624–64,896–64,896–43,264–4096–4096–1000.

Abstract

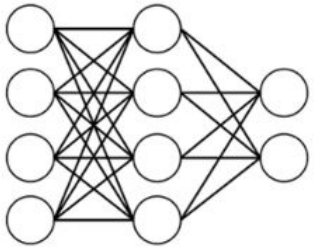




Architecture

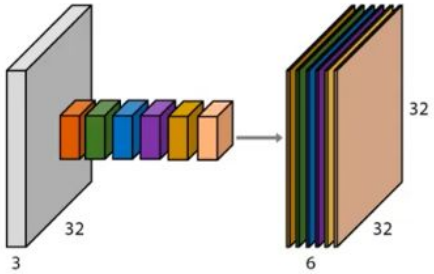


CNNs
Translation

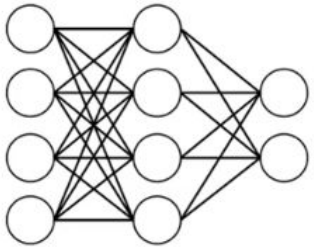


Perceptrons
Function regularity

Architecture



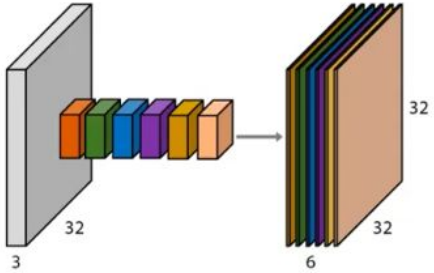
CNNs
Translation



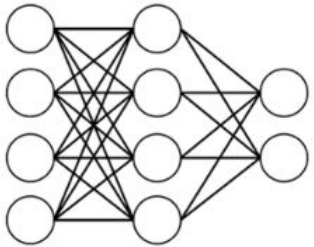
Perceptrons
Function regularity

Permutation

Architecture



CNNs
Translation

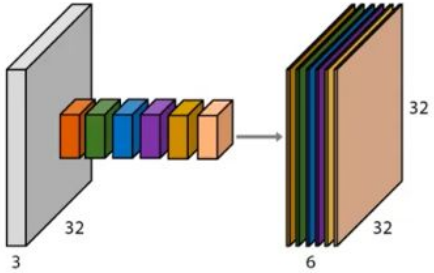


Perceptrons
Function regularity

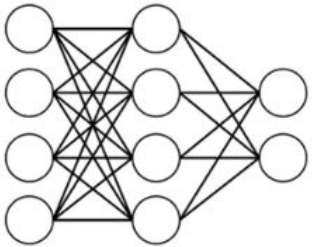


DeepSets / Transformers
Permutation

Architecture



CNNs
Translation



Perceptrons
Function regularity

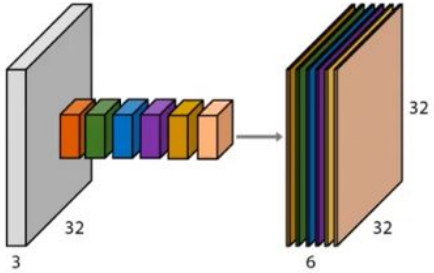


DeepSets / Transformers
Permutation



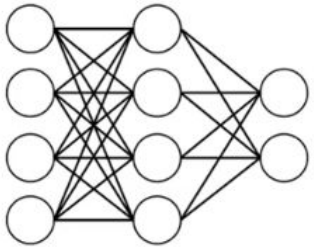
GNNs
Permutation

Architecture



CNNs
Translation

Time warping



Perceptrons
Function regularity

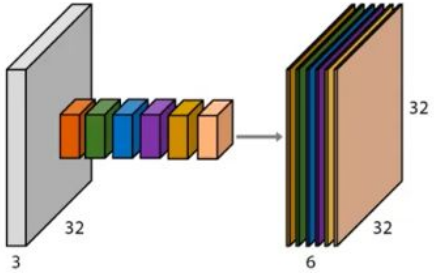


DeepSets / Transformers
Permutation

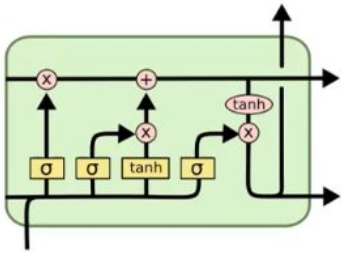


GNNs
Permutation

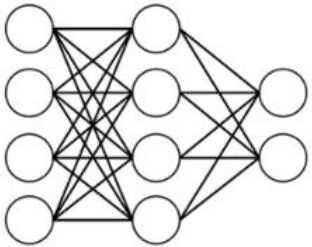
Architecture



CNNs
Translation



LSTMs
Time warping



Perceptrons
Function regularity

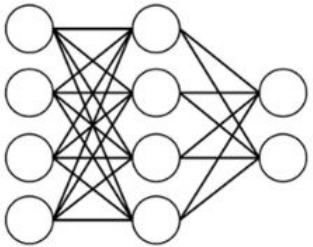


DeepSets / Transformers
Permutation

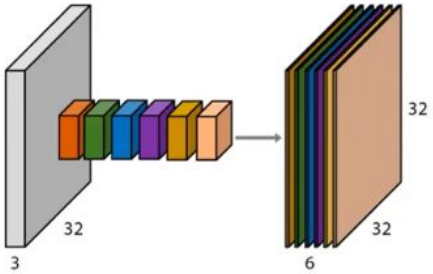


GNNs
Permutation

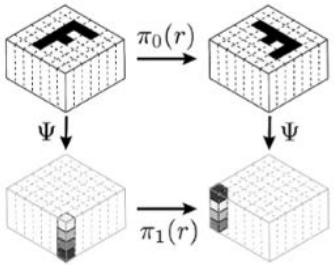
Architecture



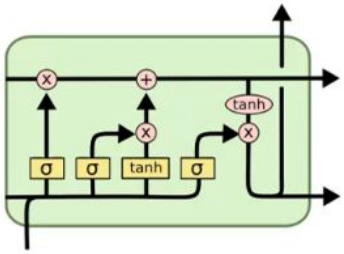
Perceptrons
Function regularity



CNNs
Translation



Group-CNNs
Translation+Rotation,
Global groups



LSTMs
Time warping

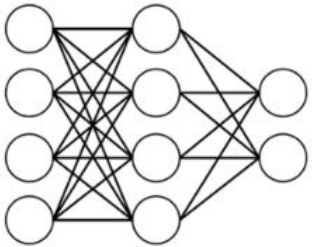


DeepSets / Transformers
Permutation

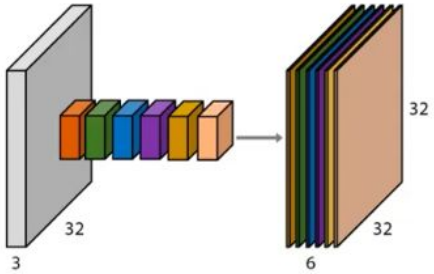


GNNs
Permutation

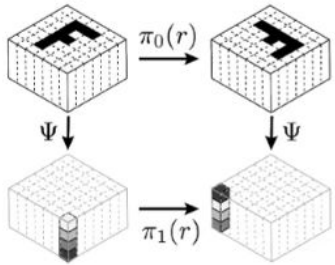
Architecture



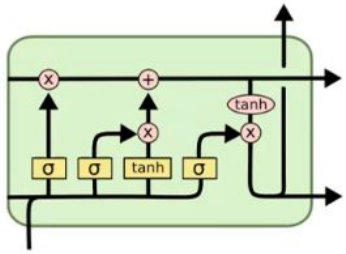
Perceptrons
Function regularity



CNNs
Translation



Group-CNNs
Translation+Rotation,
Global groups



LSTMs
Time warping



DeepSets / Transformers
Permutation



GNNs
Permutation



Intrinsic CNNs
Isometry / Gauge choice

Comparing Architectures

Table 3

Performance comparison on fifteen methods across fMRI datasets.

Method	Metric	ADNI	OASIS	PPMI	Taowu
CNN					
RNN					
LSTM					
MLP-Mixer					
Transformer					
Mamba					

Table 3

Performance comparison on fifteen methods across fMRI datasets.

Method	Metric	ADNI	OASIS	PPMI	Taowu
CNN	Acc	75.60 ± 7.88 **	88.24 ± 2.70	63.47 ± 8.02 **	80.00 ± 22.97 *
	Pre	69.23 ± 15.42 *	87.34 ± 3.41	60.95 ± 8.73	90.62 ± 14.41
	F1	68.50 ± 10.40 **	84.28 ± 3.48 **	53.68 ± 9.48 **	80.43 ± 21.77 *
RNN	Acc	71.60 ± 9.51 **	88.33 ± 2.84	64.97 ± 8.71	70.00 ± 28.38 *
	Pre	54.45 ± 13.93 **	86.52 ± 3.34	58.57 ± 16.86	89.58 ± 18.32
	F1	61.21 ± 12.18 **	85.57 ± 4.50 *	55.51 ± 11.95 *	73.43 ± 22.89 *
LSTM	Acc	74.00 ± 10.20 **	87.92 ± 2.39	66.58 ± 6.08	85.00 ± 17.48
	Pre	66.33 ± 18.51 *	84.33 ± 3.45	59.00 ± 11.95	90.83 ± 23.72
	F1	67.17 ± 14.31 **	83.05 ± 3.33 *	57.15 ± 8.35 *	86.38 ± 20.85
MLP-Mixer	Acc	72.80 ± 9.00 **	87.51 ± 2.51 *	66.05 ± 7.33 *	87.50 ± 17.68
	Pre	68.32 ± 11.49 *	82.68 ± 4.33	64.90 ± 8.57	91.25 ± 18.45
	F1	64.11 ± 10.80 **	81.99 ± 3.21 **	62.05 ± 7.58 **	88.10 ± 16.49
Transformer	Acc	74.00 ± 7.60 **	87.84 ± 2.20 **	63.97 ± 9.32 *	87.50 ± 17.68
	Pre	66.67 ± 12.96 *	83.58 ± 3.83	57.80 ± 11.78	82.08 ± 26.92
	F1	65.84 ± 8.60 **	83.48 ± 1.64 **	55.06 ± 12.30 **	83.52 ± 23.55
Mamba	Acc	79.20 ± 4.66	88.33 ± 1.99 *	64.21 ± 3.94	92.50 ± 12.08
	Pre	78.19 ± 8.88	85.83 ± 2.75	59.91 ± 11.65	94.37 ± 13.24
	F1	75.31 ± 5.49	84.72 ± 2.81 *	59.91 ± 7.45 *	92.67 ± 12.19

Table 3

Performance

Method					
CNN	GCN	Acc	74.80 ± 7.55 **	87.67 ± 2.62 *	66.00 ± 9.33 *	85.00 ± 24.15
		Pre	71.16 ± 11.19 *	84.65 ± 3.01	62.81 ± 9.80	**97.08 ± 6.23**
		F1	68.29 ± 8.76 **	83.33 ± 3.52 *	57.05 ± 11.53 **	87.57 ± 19.76
	GIN	Acc	78.00 ± 7.36	87.43 ± 3.10 *	67.00 ± 10.56	92.50 ± 12.08
		Pre	79.66 ± 6.81	81.69 ± 7.94	59.84 ± 14.76	95.63 ± 13.84
		F1	73.00 ± 9.32 *	82.18 ± 4.61 *	60.30 ± 11.45 *	93.57 ± 11.88
	MGNN	Acc	75.60 ± 8.10 **	87.76 ± 2.65	69.66 ± 7.35	**97.50 ± 7.91**
		Pre	75.09 ± 9.38	86.53 ± 4.23	62.09 ± 10.02	95.63 ± 13.83
		F1	70.12 ± 8.98 **	82.78 ± 3.31	62.46 ± 8.02 **	96.43 ± 11.29
RNN	SPDNet	Acc	78.40 ± 9.08	89.14 ± 2.97	64.97 ± 9.99 *	77.50 ± 18.45 *
		Pre	79.87 ± 9.44	88.09 ± 3.33	58.31 ± 16.37 *	95.83 ± 6.80
		F1	74.08 ± 11.34 *	86.66 ± 3.95	55.67 ± 13.14 **	83.14 ± 13.29 *
LSTM	RResNet	Acc	74.40 ± 8.62	88.81 ± 2.13	66.47 ± 10.66	85.00 ± 12.25
		Pre	67.16 ± 14.09	87.74 ± 2.97	58.04 ± 10.19	79.58 ± 19.80
		F1	67.09 ± 11.51	84.87 ± 3.06	57.11 ± 13.39	80.71 ± 16.24
	BrainGNN	Acc	79.20 ± 5.00	87.25 ± 1.42	67.18 ± 6.85	80.00 ± 10.00 *
		Pre	79.50 ± 10.20	76.15 ± 2.49 *	61.49 ± 11.35	82.08 ± 14.29
		F1	75.12 ± 7.29	81.32 ± 2.04 *	59.69 ± 8.80 *	77.52 ± 11.97 *
MLP-Mixer	ContrastPool	Acc	76.00 ± 6.93 *	87.27 ± 2.72 *	67.50 ± 9.16	82.50 ± 16.01 *
		Pre	57.2 ± 19.84 *	50.33 ± 14.27 *	42.71 ± 19.82 *	75.00 ± 28.32 *
		F1	56.19 ± 13.54 *	48.07 ± 3.48 **	39.55 ± 16.68 **	73.90 ± 25.10 *
Transformer	NeuroGraph	Acc	76.40 ± 7.41 *	87.13 ± 3.00	69.08 ± 9.02 *	87.50 ± 17.68 *
		Pre	80.02 ± 5.90	**88.86 ± 2.22**	**79.13 ± 3.07** **	89.58 ± 16.32
		F1	71.52 ± 8.82 *	81.16 ± 4.29 *	61.12 ± 11.43 **	85.19 ± 19.91 *
	STAGIN	Acc	77.60 ± 6.31	88.56 ± 0.78	69.66 ± 5.71	90.00 ± 12.91
		Pre	62.75 ± 38.97	83.21 ± 17.41	69.66 ± 5.71	90.00 ± 16.10
		F1	62.71 ± 27.06 *	83.28 ± 10.13 *	**69.66 ± 5.71**	90.67 ± 12.65
Mamba	*Geo-Mamba*	Acc	**81.20 ± 6.21**	**89.39 ± 1.91**	**71.16 ± 9.12**	**97.50 ± 7.50**
		Pre	**81.10 ± 8.12**	88.82 ± 2.32	64.81 ± 10.10	95.63 ± 13.13
		F1	**79.08 ± 7.43**	**87.10 ± 2.57**	65.57 ± 10.17	**96.43 ± 10.71**

Comparing Architectures

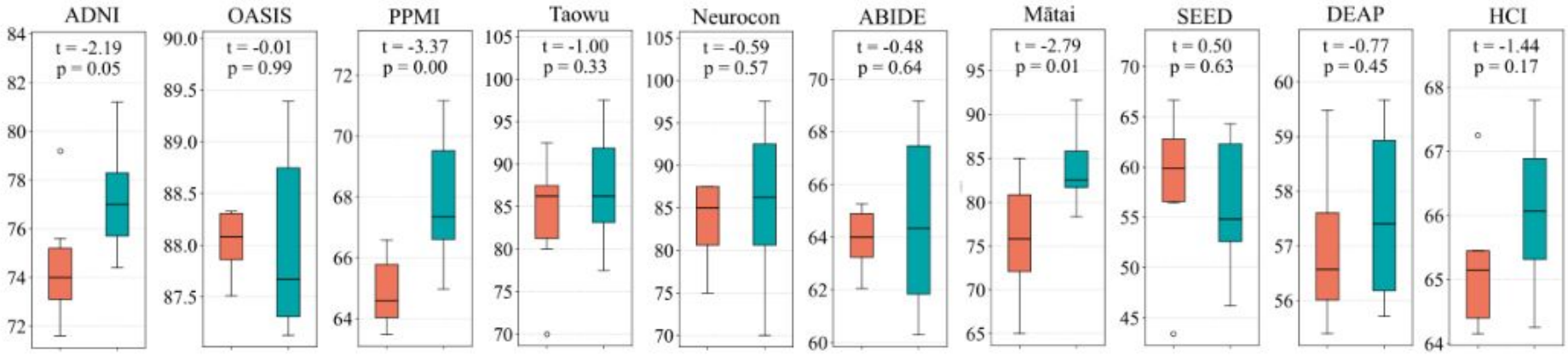
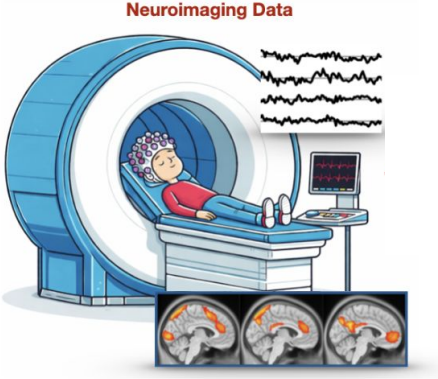


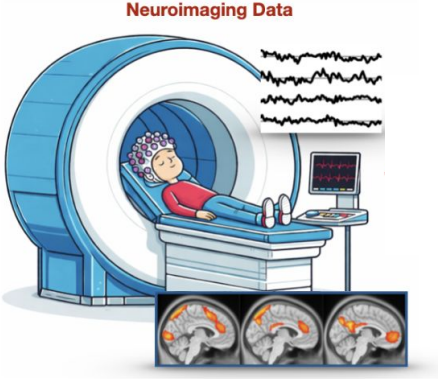
Fig. 2. The overall performance between sequential models ( ) and spatial models ( ) for various datasets.

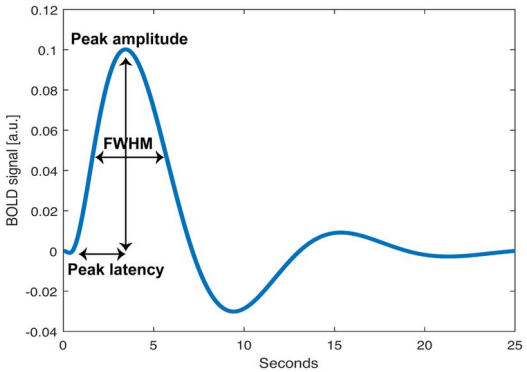
Geo-Mamba Architecture

fMRI BOLD Time Series

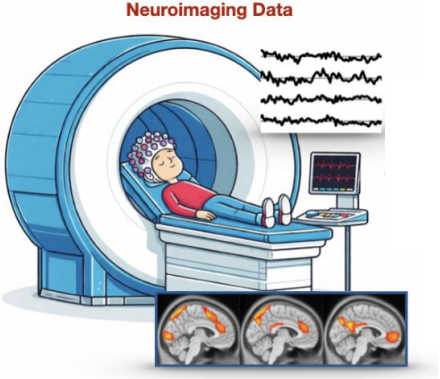


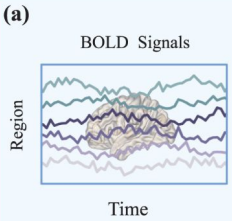
fMRI BOLD Time Series

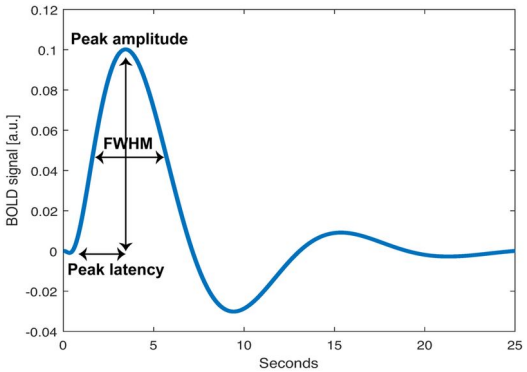




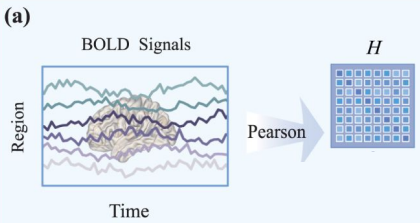
fMRI BOLD Time Series



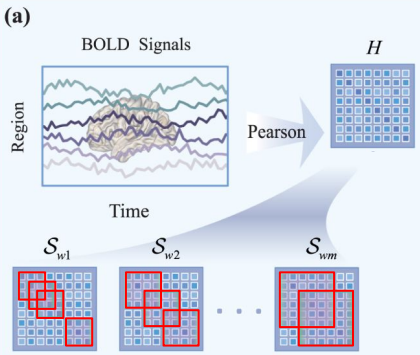


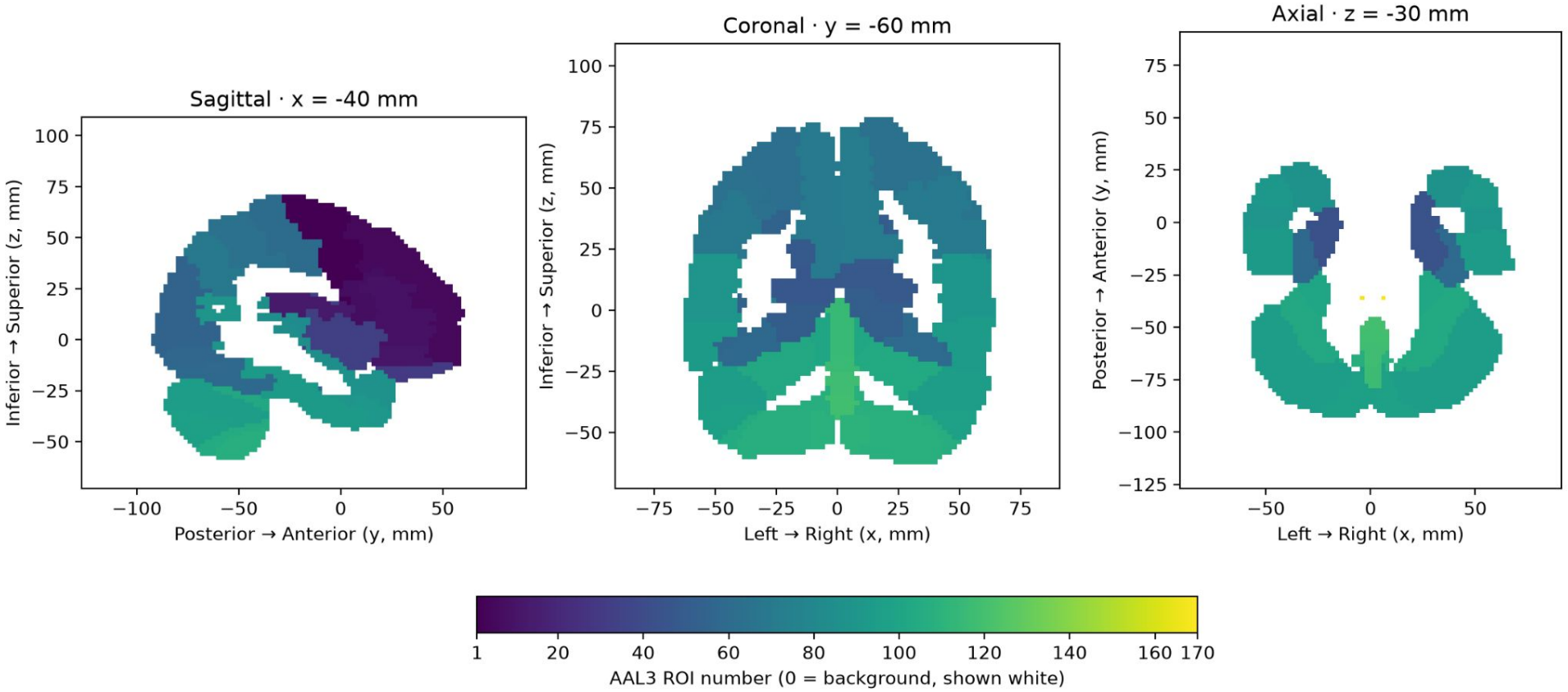


Functional Connectivity Matrix

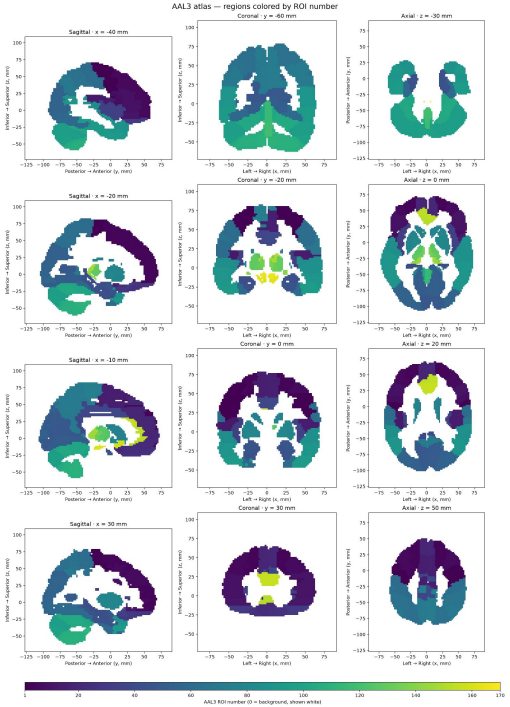


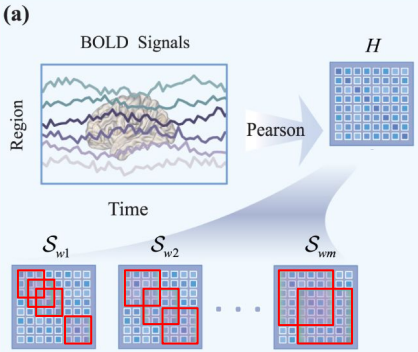
Gather Block Diagonals




AAL3 atlas — regions colored by ROI number
Sagittal · x = -40 mm
Inferior → Superior (z, mm)
Posterior → Anterior (y, mm)
Coronal · y = -60 mm
Inferior → Superior (z, mm)
Left → Right (x, mm)
Axial · z = -30 mm
Posterior → Anterior (y, mm)
Left → Right (x, mm)
AAL3 ROI number (0 = background, shown white)

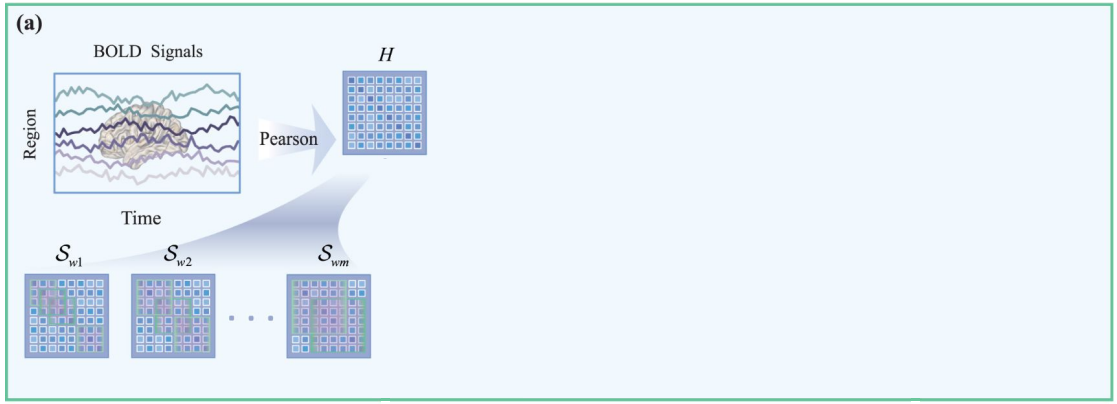
Gather Block Diagonals





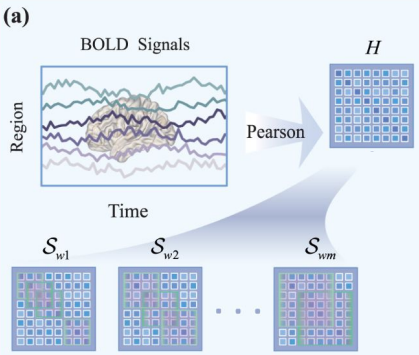
Properties of Covariance Matrices

$$\Sigma = \frac{1}{m-1} X X^\top$$



Properties of Covariance Matrices

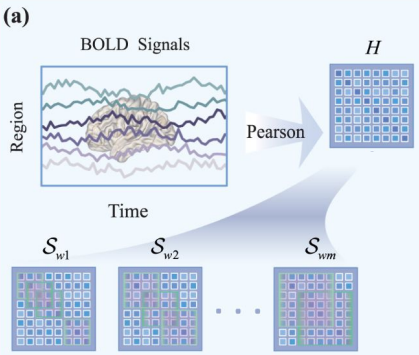
$$\Sigma = \frac{1}{m-1} XX^\top$$



$$\mathrm{Cov}(AX) = \frac{1}{m-1}(AX)(AX)^\top$$

Properties of Covariance Matrices

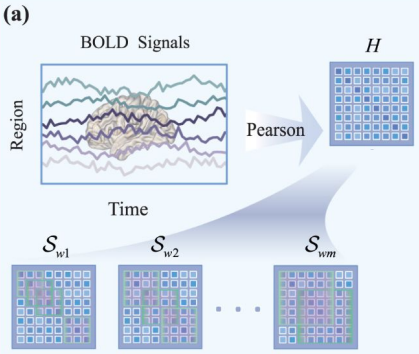
$$\Sigma = \frac{1}{m-1} XX^\top$$



$$\begin{aligned}
\mathrm{Cov}(AX) &= \frac{1}{m-1}(AX)(AX)^\top \\
&= \frac{1}{m-1} AXX^\top A^\top \\
&= A\left(\frac{1}{m-1} XX^\top\right) A^\top
\end{aligned}$$

Properties of Covariance Matrices

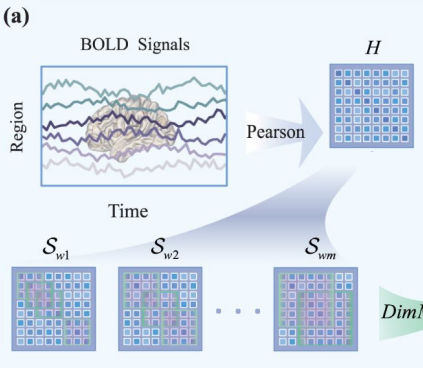
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&= \boxed{A\,\mathrm{Cov}(X)A^\top}
\end{aligned}$$

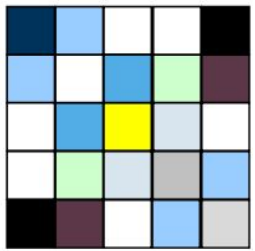
Properties of Covariance Matrices

$$\Sigma = \frac{1}{m-1} XX^\top$$



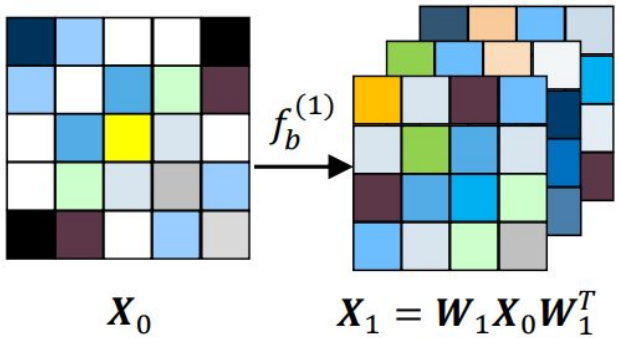
$$\begin{aligned}
\mathrm{Cov}(AX) &= \frac{1}{m-1}(AX)(AX)^\top \\
&= \frac{1}{m-1} AXX^\top A^\top \\
&= A\left(\frac{1}{m-1} XX^\top\right) A^\top \\
&= \boxed{A\,\mathrm{Cov}(X)A^\top}
\end{aligned}$$

Properties of Covariance Matrices



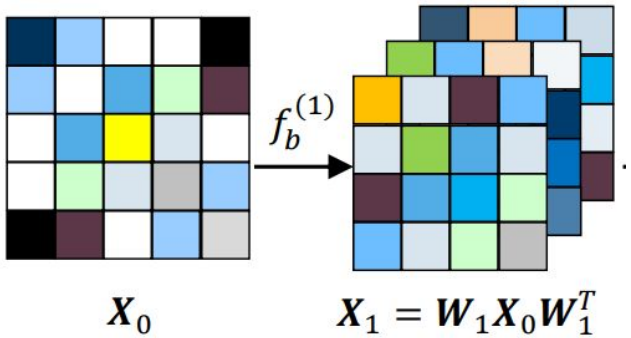
X_0

Properties of Covariance Matrices



Properties of Covariance Matrices

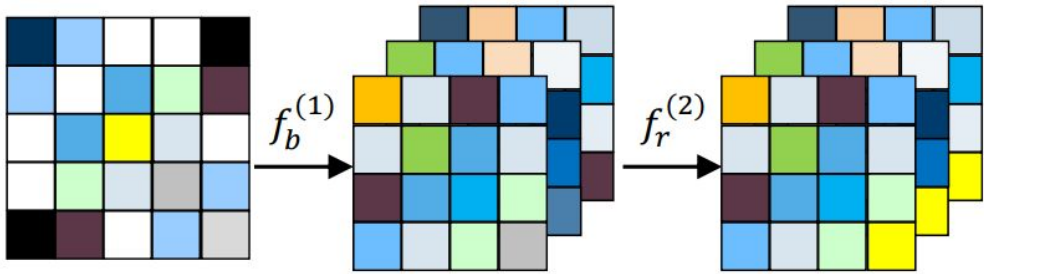
Convolution-Like Layer



Properties of Covariance Matrices

Convolution-Like Layer

ReLU-Like Layer



$\boldsymbol{X}_0$

$\boldsymbol{X}_1 = \boldsymbol{W}_1 \boldsymbol{X}_0 \boldsymbol{W}_1^T$

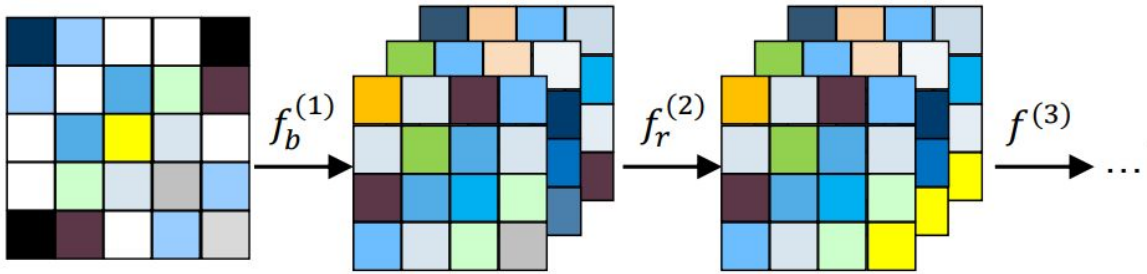
$\boldsymbol{X}_2 = \boldsymbol{U}_1 \max(\epsilon \boldsymbol{I}, \boldsymbol{\Sigma}_1) \boldsymbol{U}_1^T$

$\boldsymbol{X}_1 = \boldsymbol{U}_1 \boldsymbol{\Sigma}_1 \boldsymbol{U}_1^T$

Properties of Covariance Matrices

Convolution-Like Layer

ReLU-Like Layer



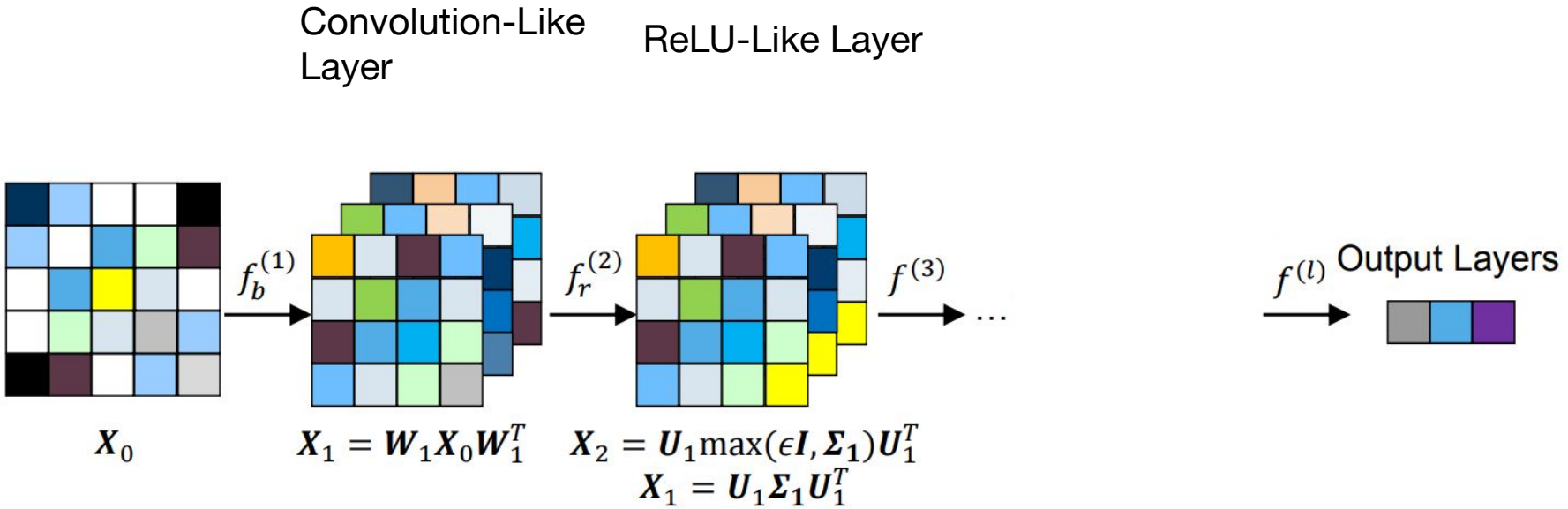
$\boldsymbol{X_0}$

$\boldsymbol{X_1 = W_1 X_0 W_1^T}$

$\boldsymbol{X_2 = U_1 \max(\epsilon I, \Sigma_1) U_1^T}$

$\boldsymbol{X_1 = U_1 \Sigma_1 U_1^T}$

Properties of Covariance Matrices



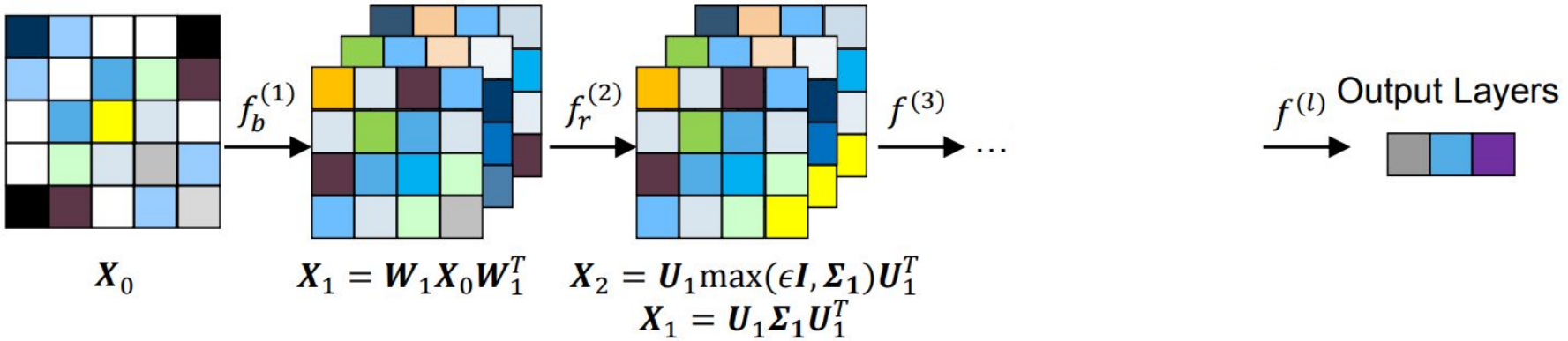
SPDNet

A Riemannian Network for SPD Matrix Learning

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{zhiwu.huang, vangool}@vision.ee.ethz.ch

Convolution-Like Layer

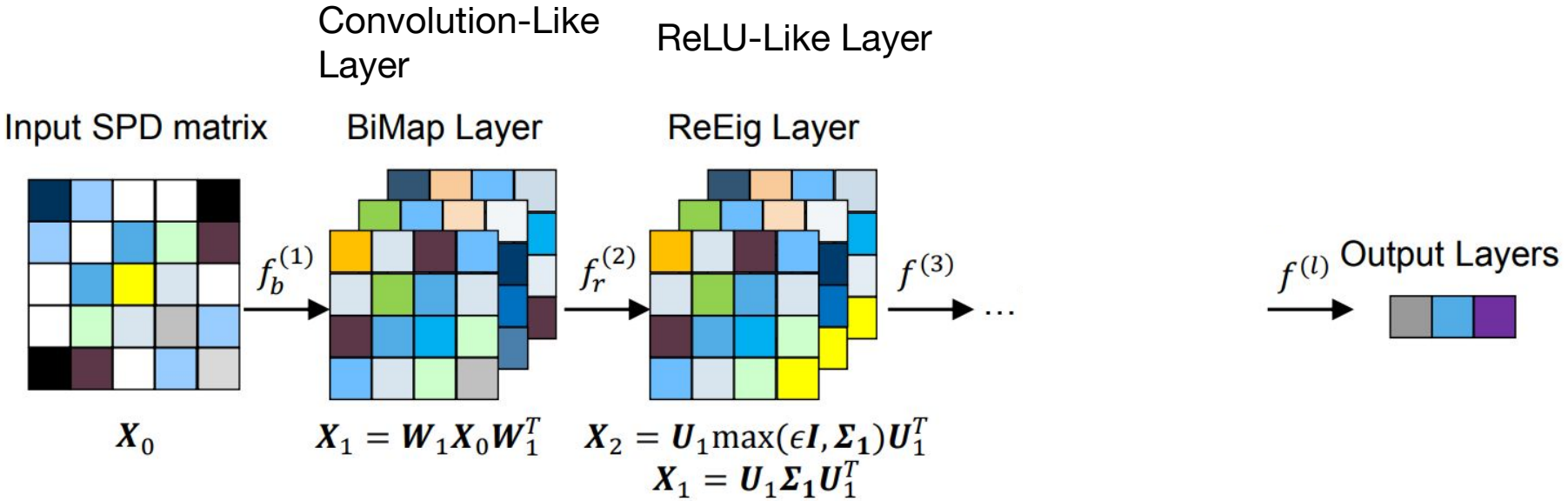
ReLU-Like Layer



SPDNet

A Riemannian Network for SPD Matrix Learning

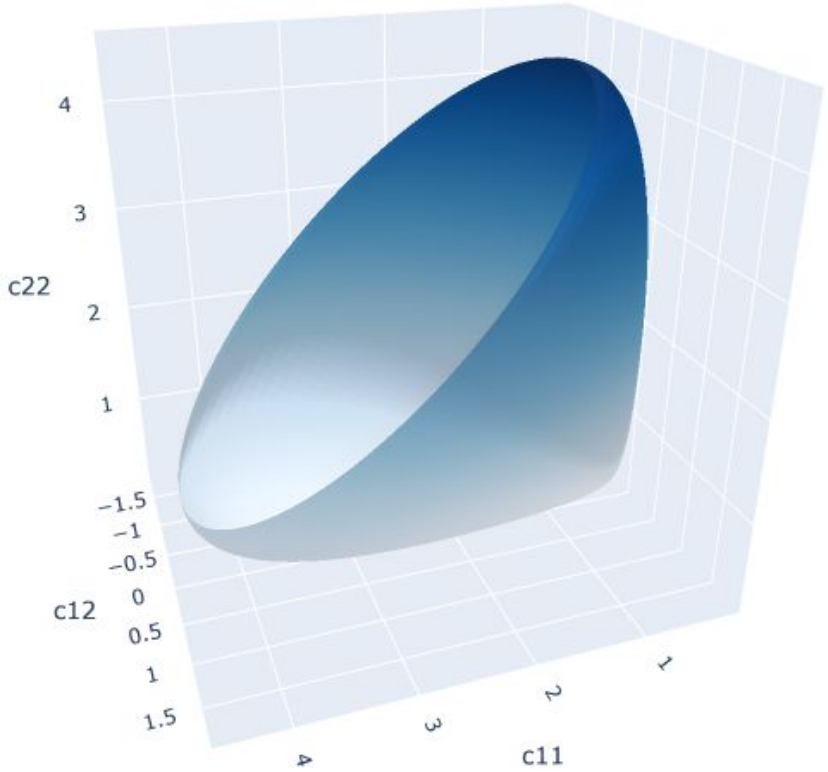
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$$\mathcal{S}_{++}^{n}$$

$$X \in \mathbb{R}^{n \times n}$$

$$X = X^{\top}$$

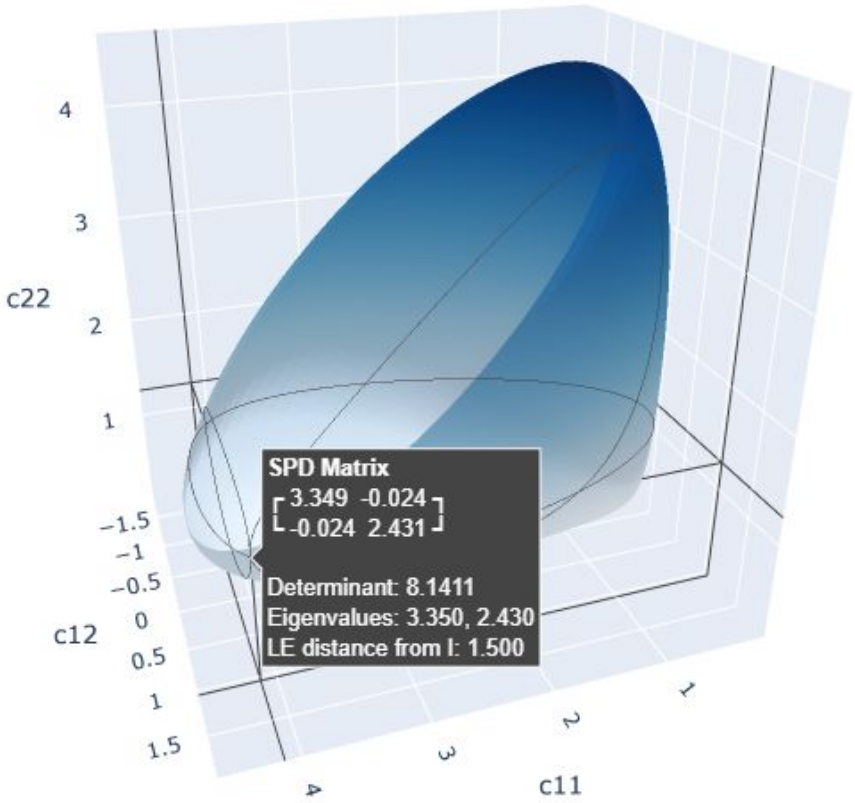
$$X \succ 0$$



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$$\mathcal{S}_{++}^{n}$$

$$X \in \mathbb{R}^{n \times n}$$

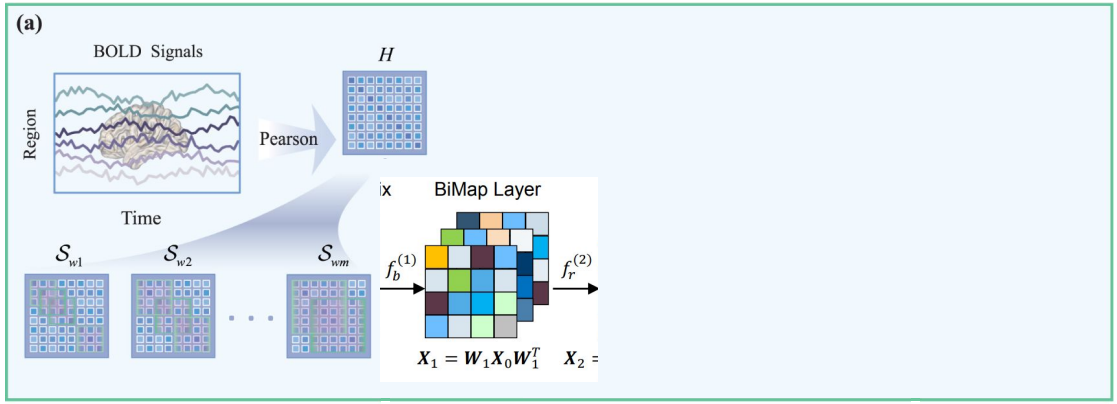
$$X = X^{\top}$$

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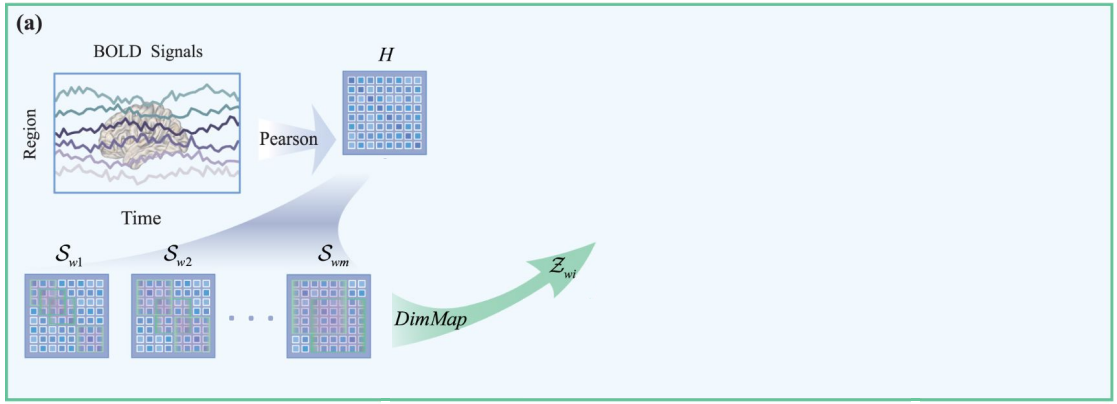


[SPDLearn Examples]

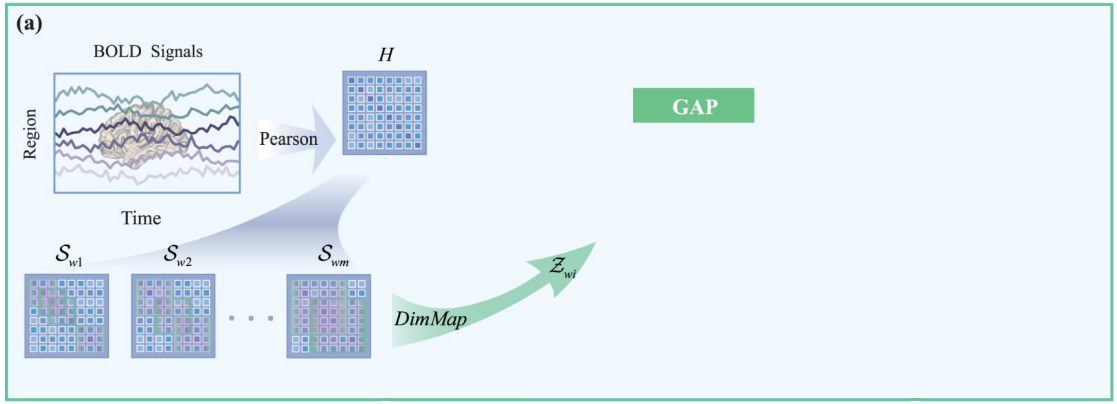
BiMap/DimMap



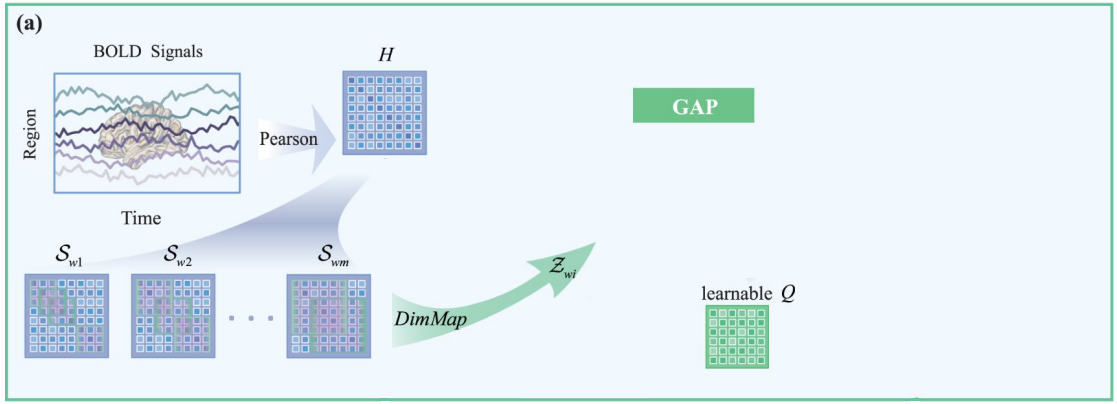
BiMap/DimMap



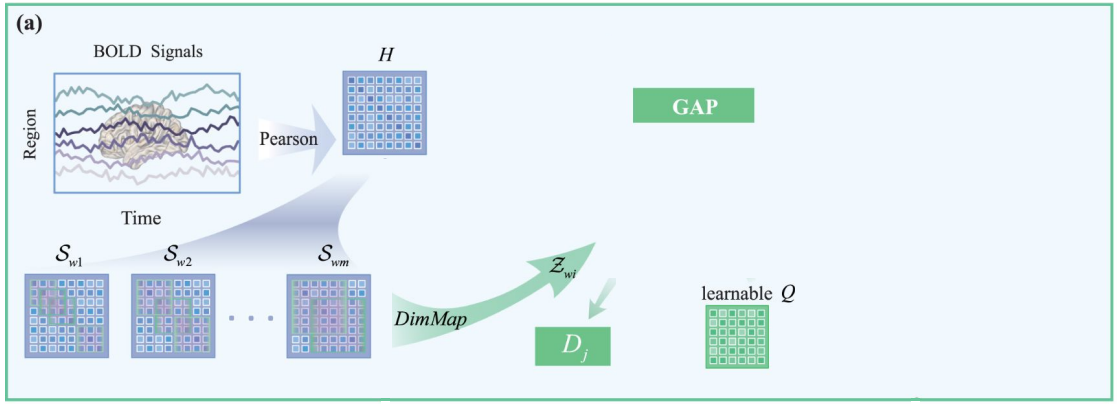
Geometric Attention Pooling



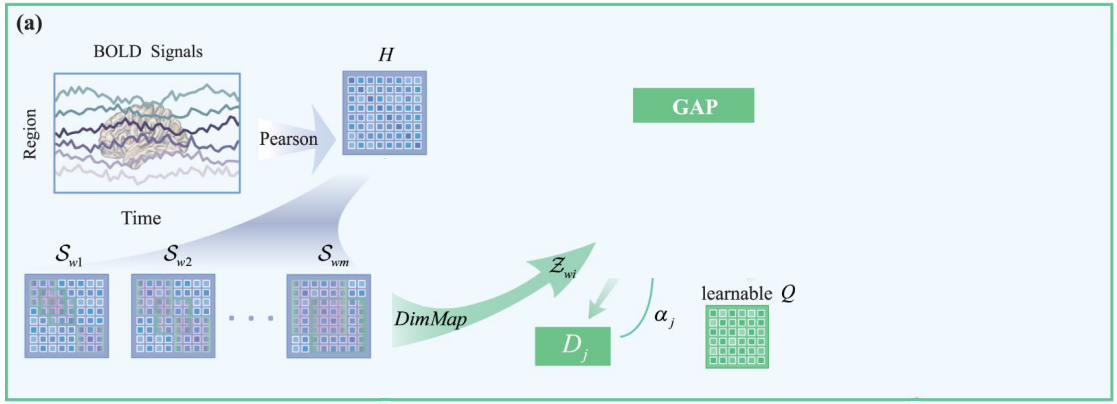
Geometric Attention Pooling



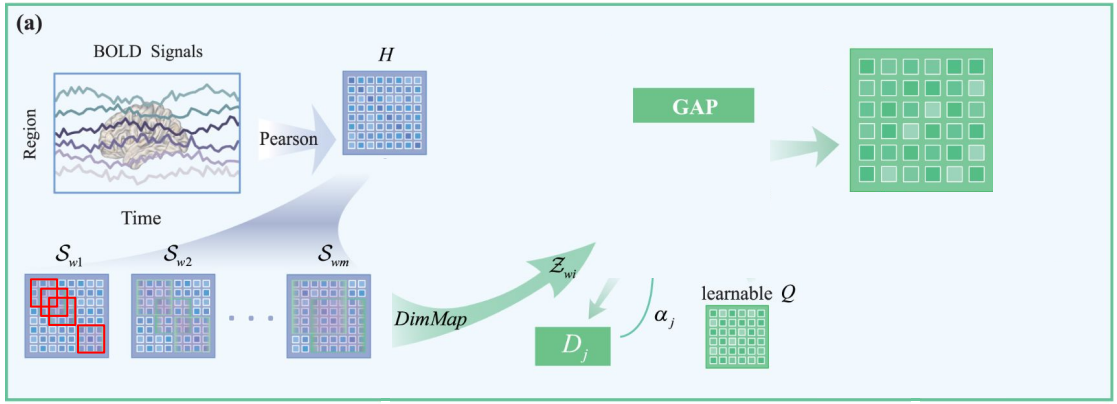
Geometric Attention Pooling



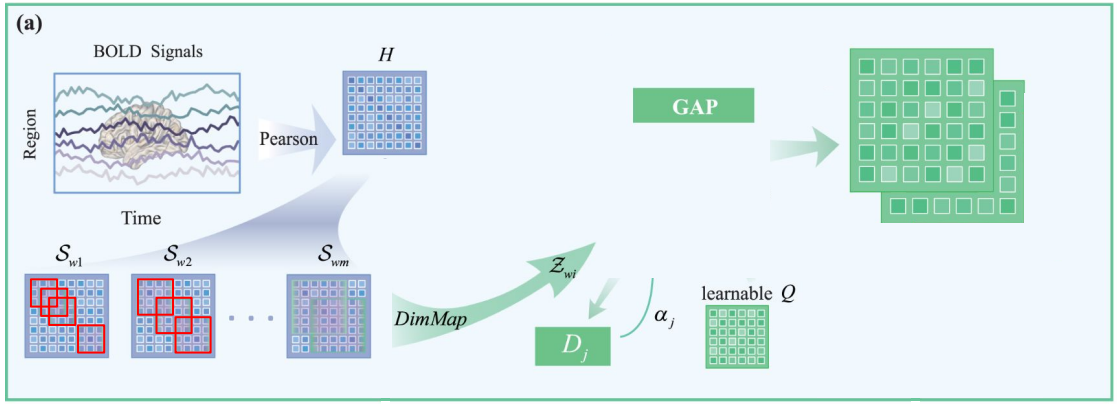
Geometric Attention Pooling



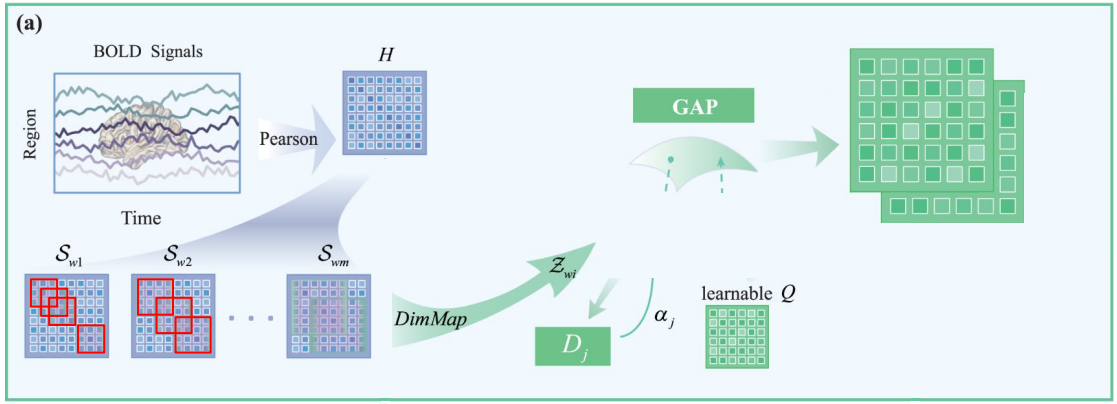
Geometric Attention Pooling



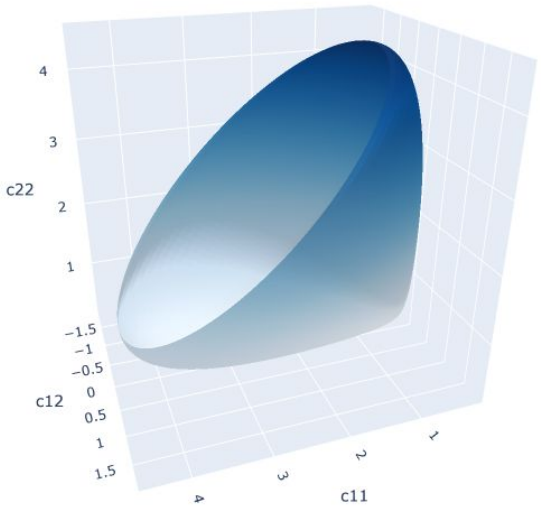
Geometric Attention Pooling

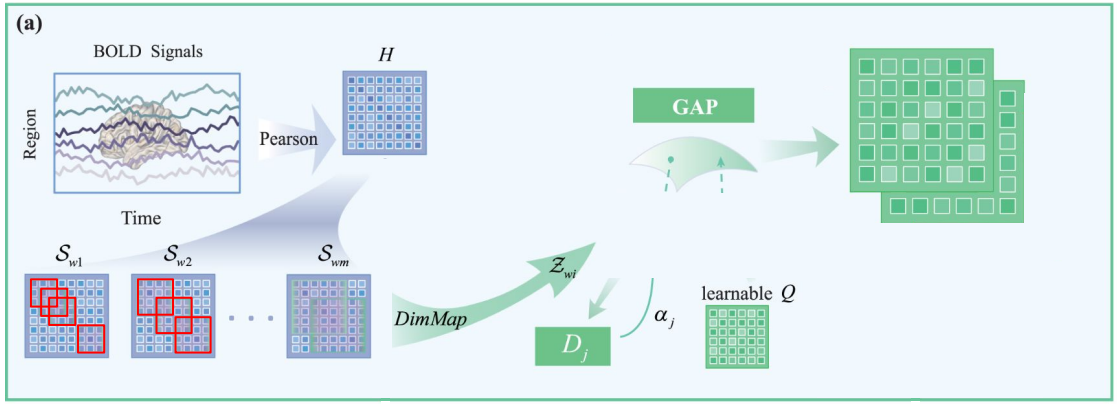


Geometric Attention Pooling

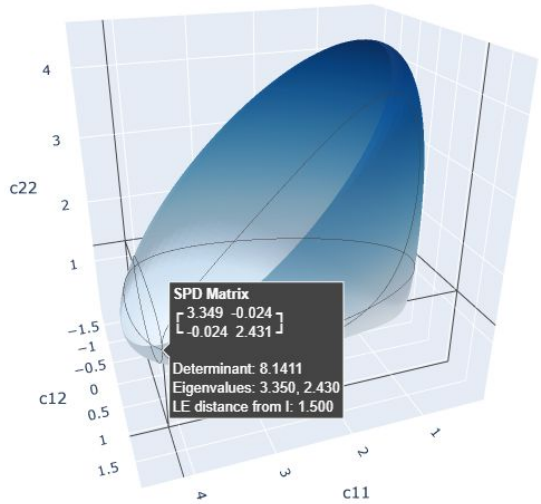


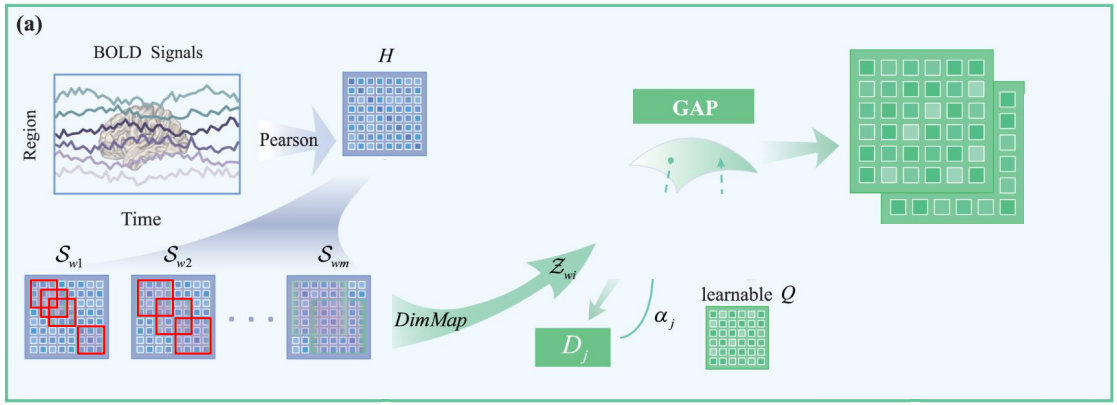
Geometric Attention Pooling



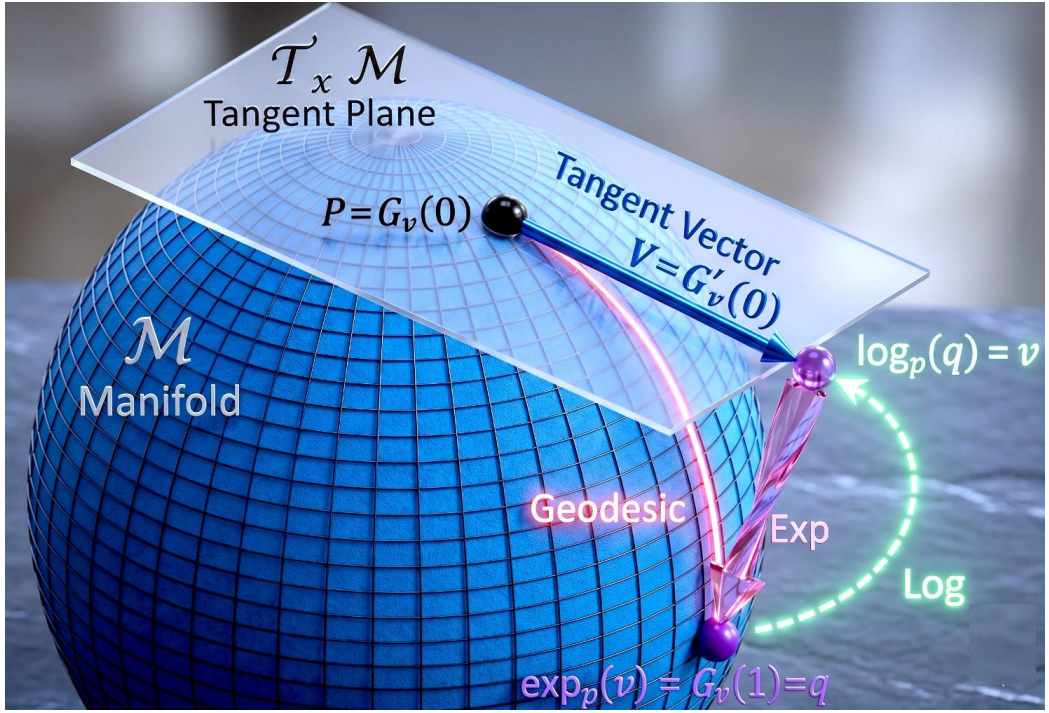


Geometric Attention Pooling





Geometric Attention Pooling

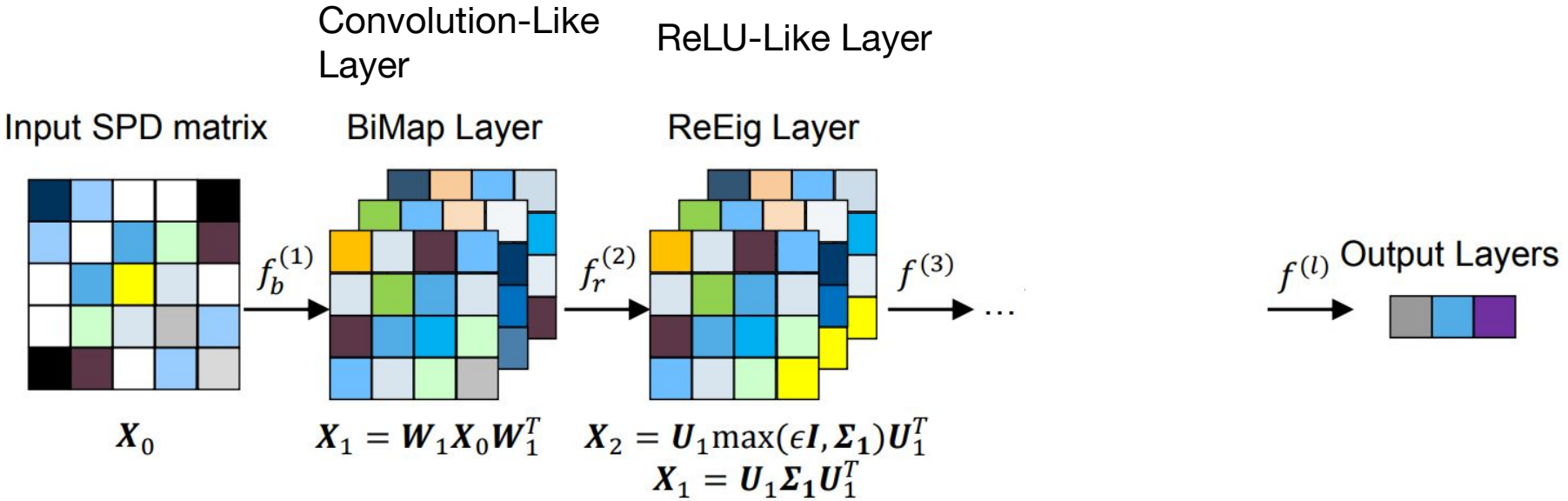


https://patricknicolas.substack.com/p/a-friendly-primer-on-geometric-deep?open=false#%C2%A7manifold-and-group-based-learning

SPDNet

A Riemannian Network for SPD Matrix Learning

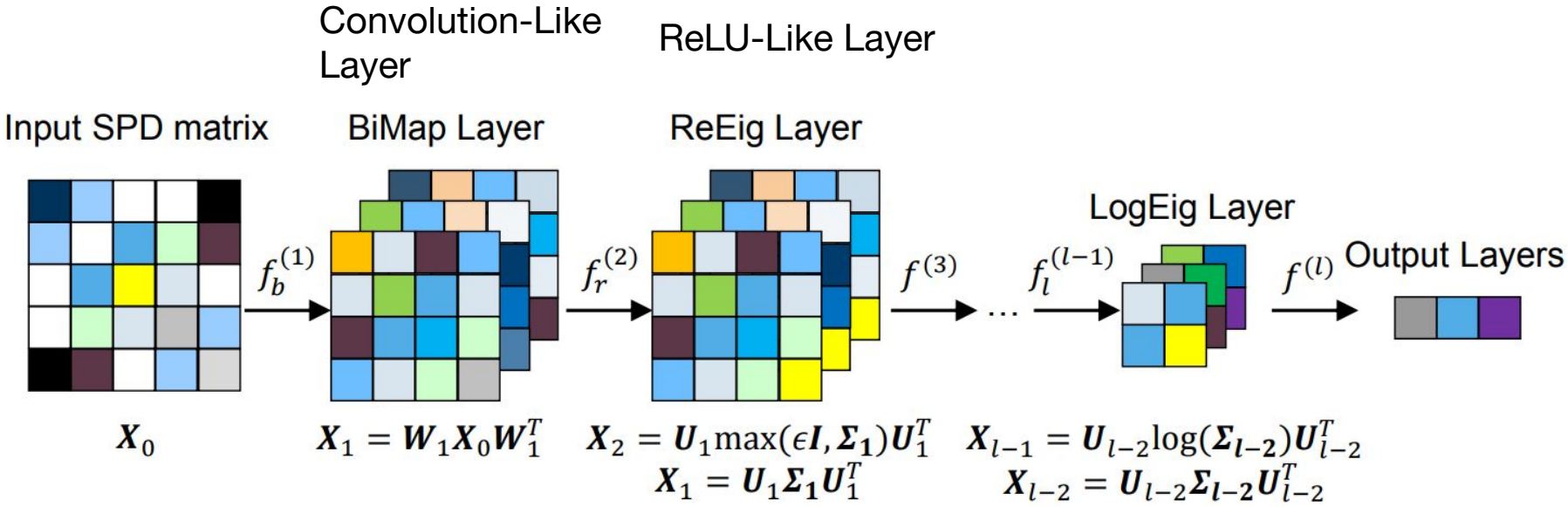
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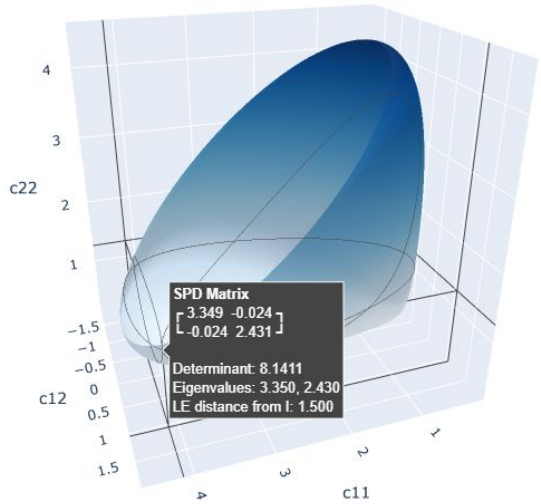
SPDNet

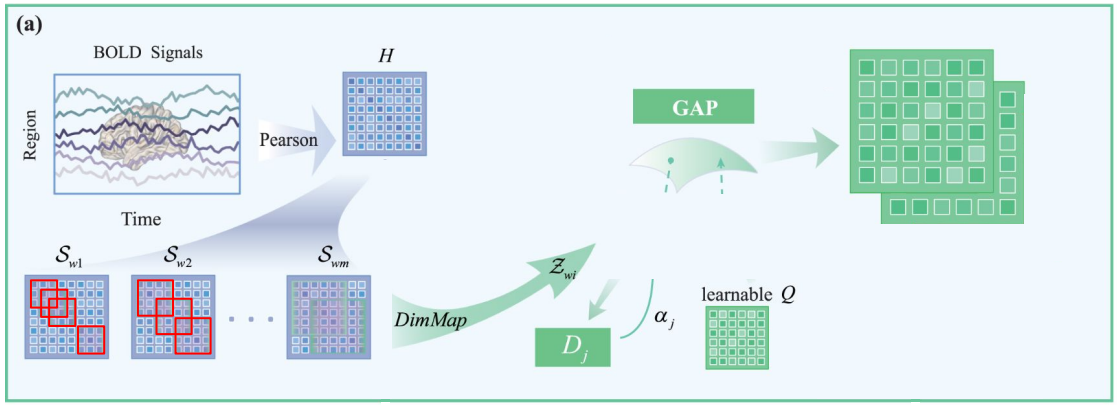
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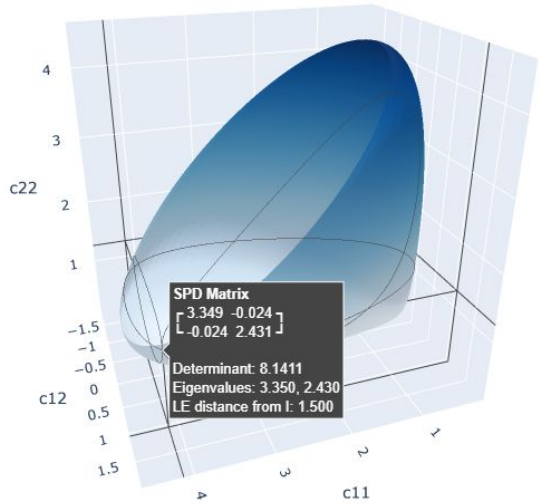


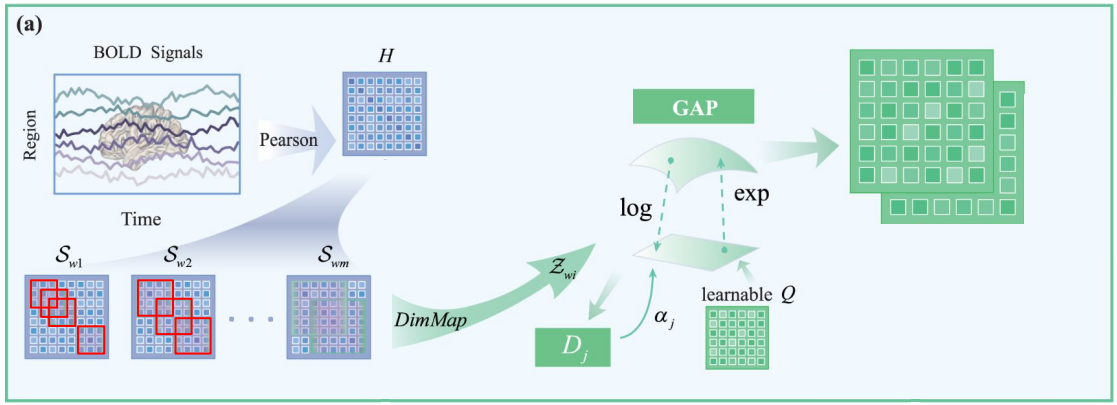
Geometric Attention Pooling



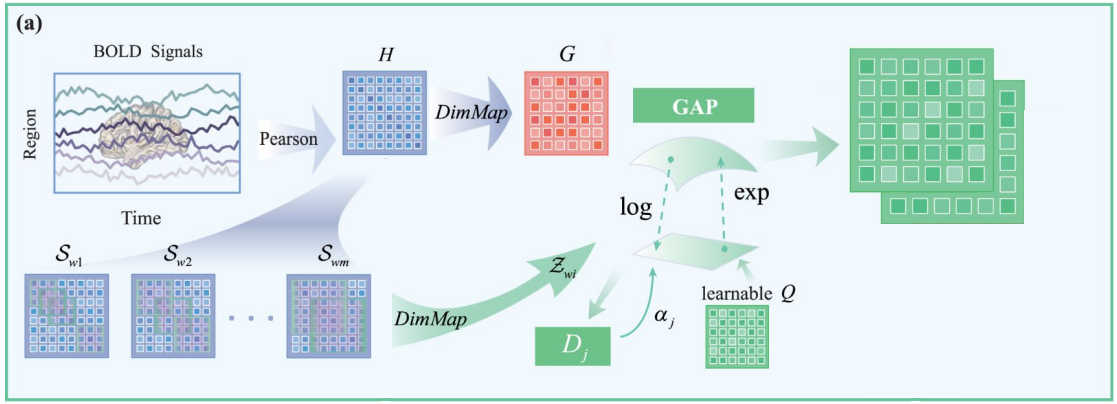


Geometric Attention Pooling

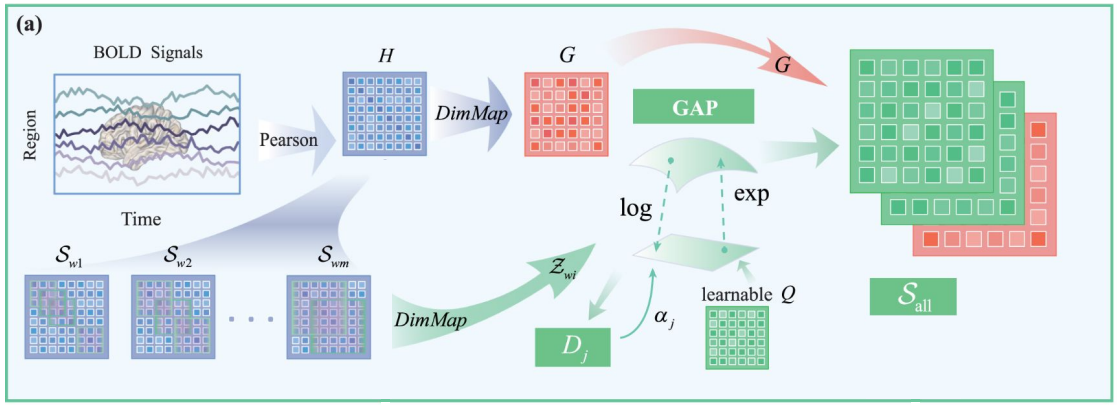




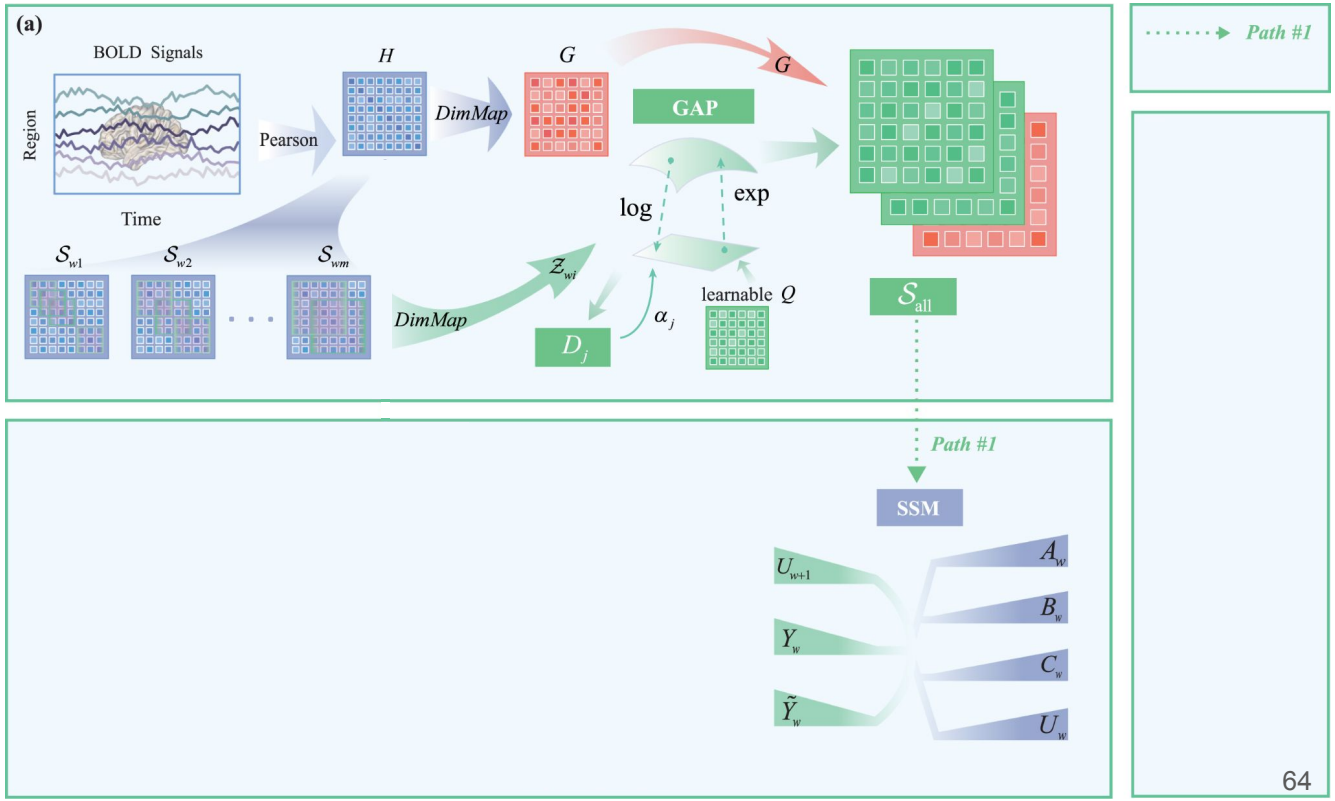
Global Token G



Local-to-Global Pyramid



Pseudo-Temporal Sequence Transduction



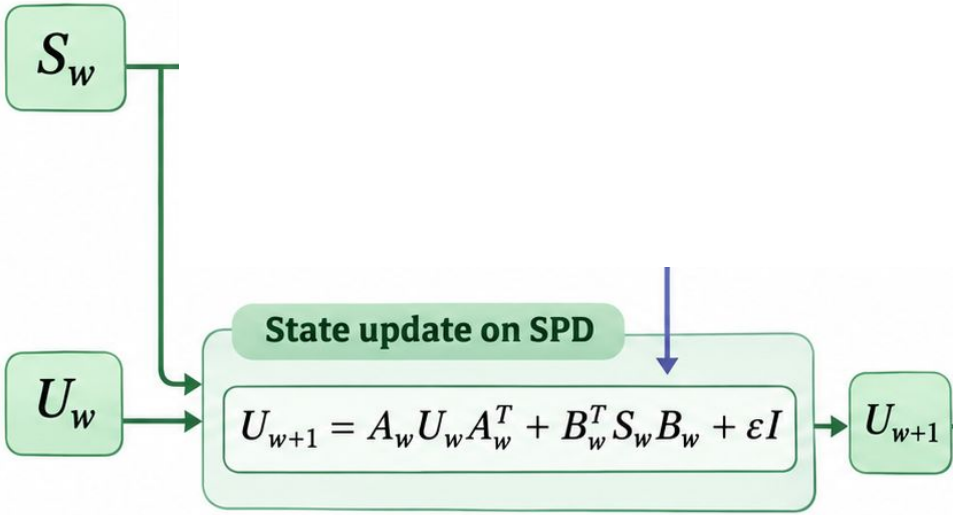
Selective R-SSM block on the SPD manifold

Selective R-SSM block on the SPD manifold

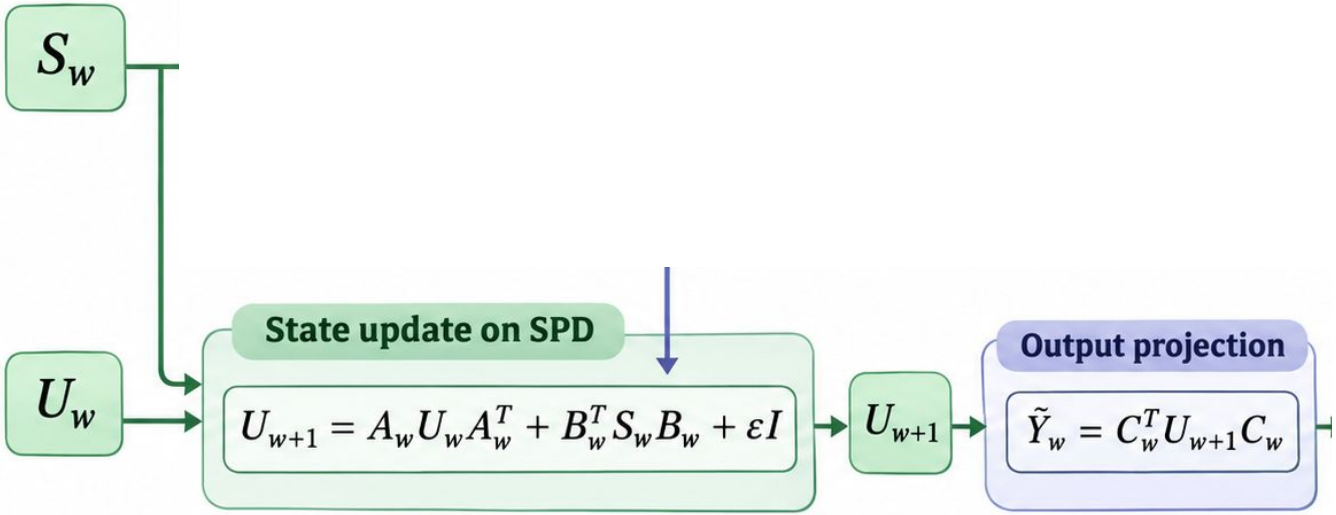
S_w

U_w

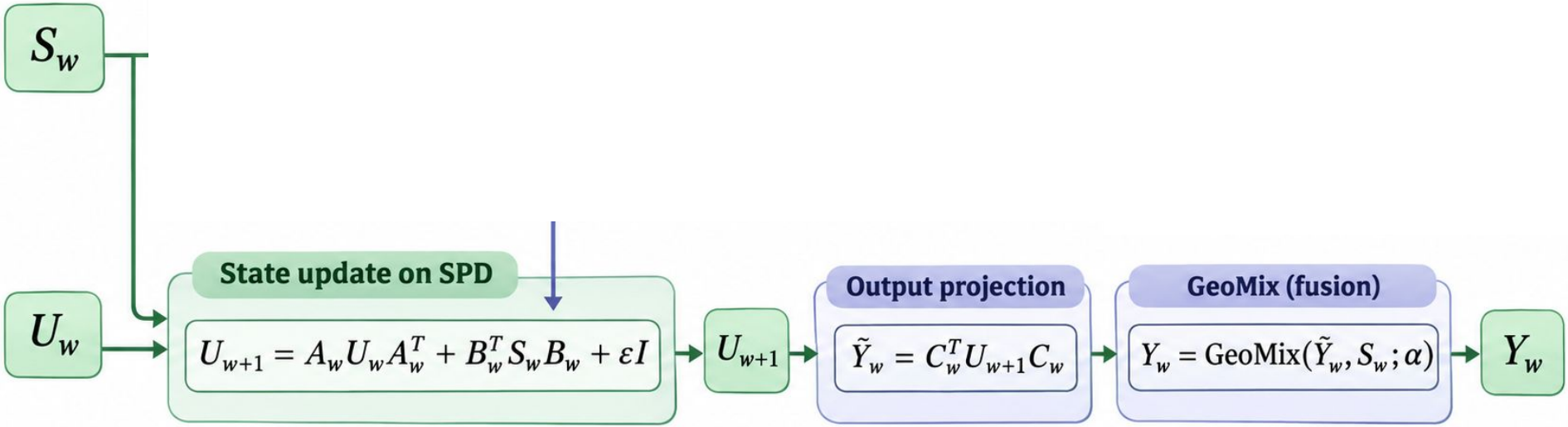
Selective R-SSM block on the SPD manifold



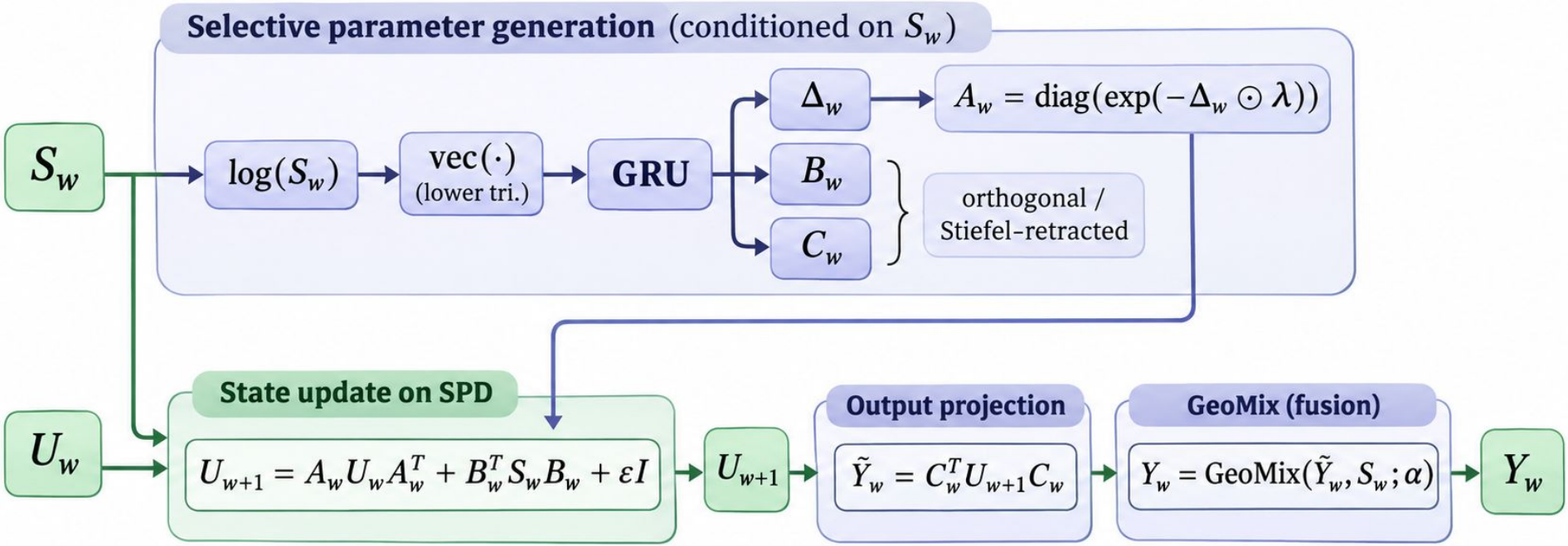
Selective R-SSM block on the SPD manifold



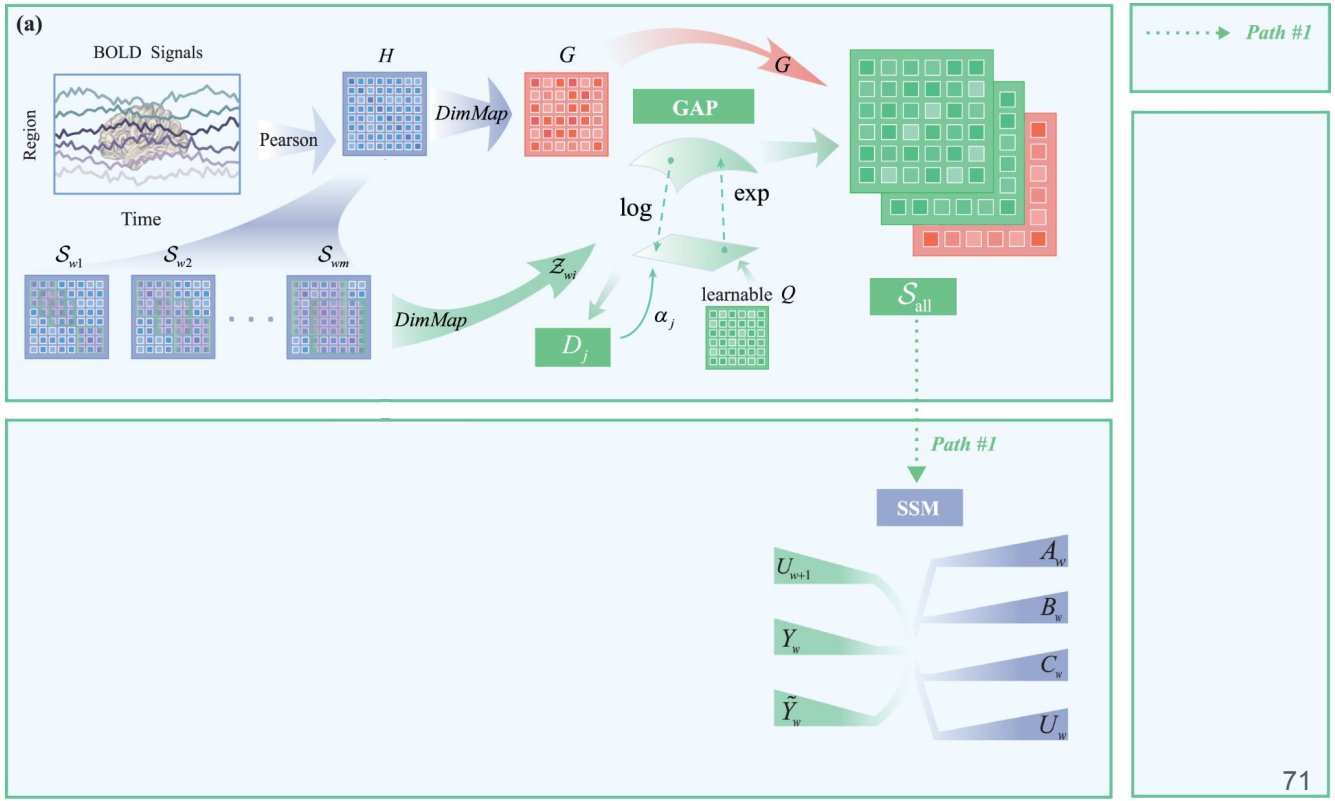
Selective R-SSM block on the SPD manifold



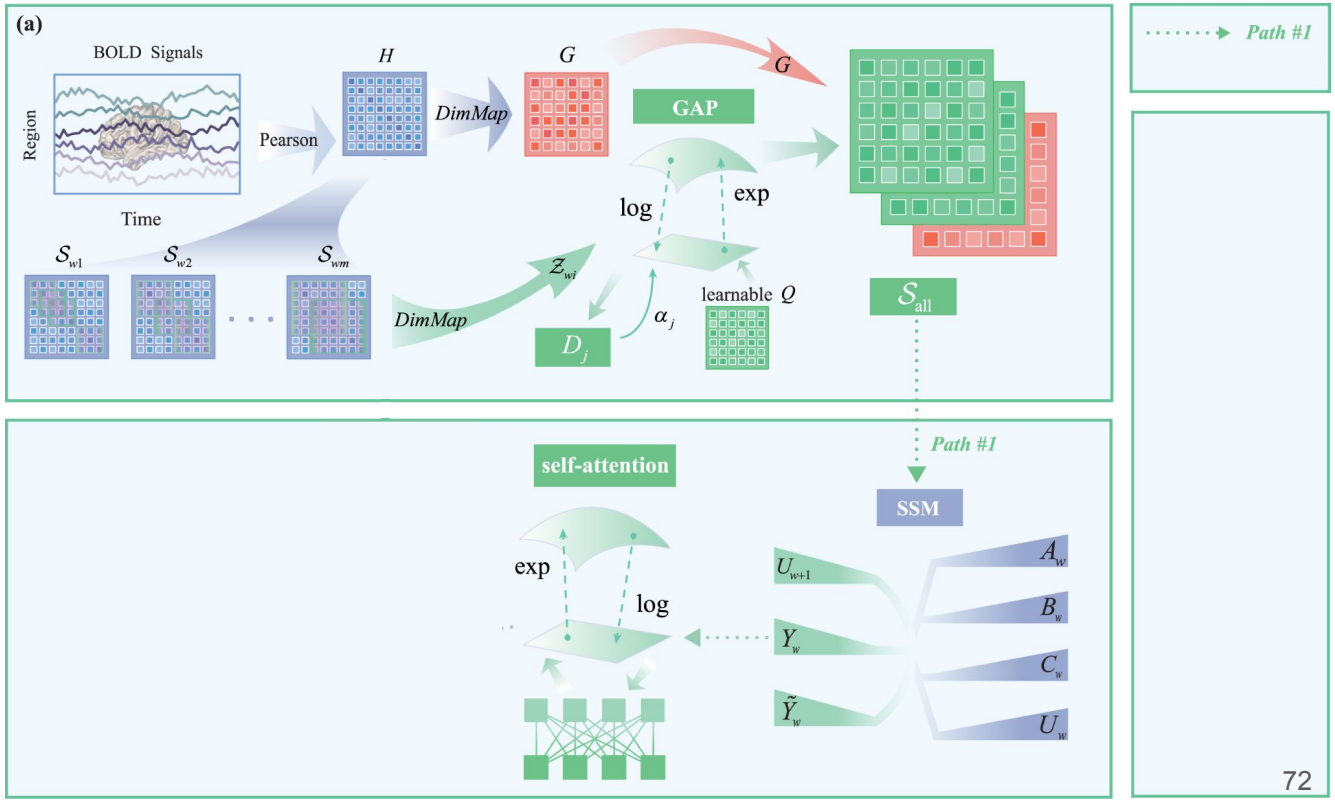
Selective R-SSM block on the SPD manifold



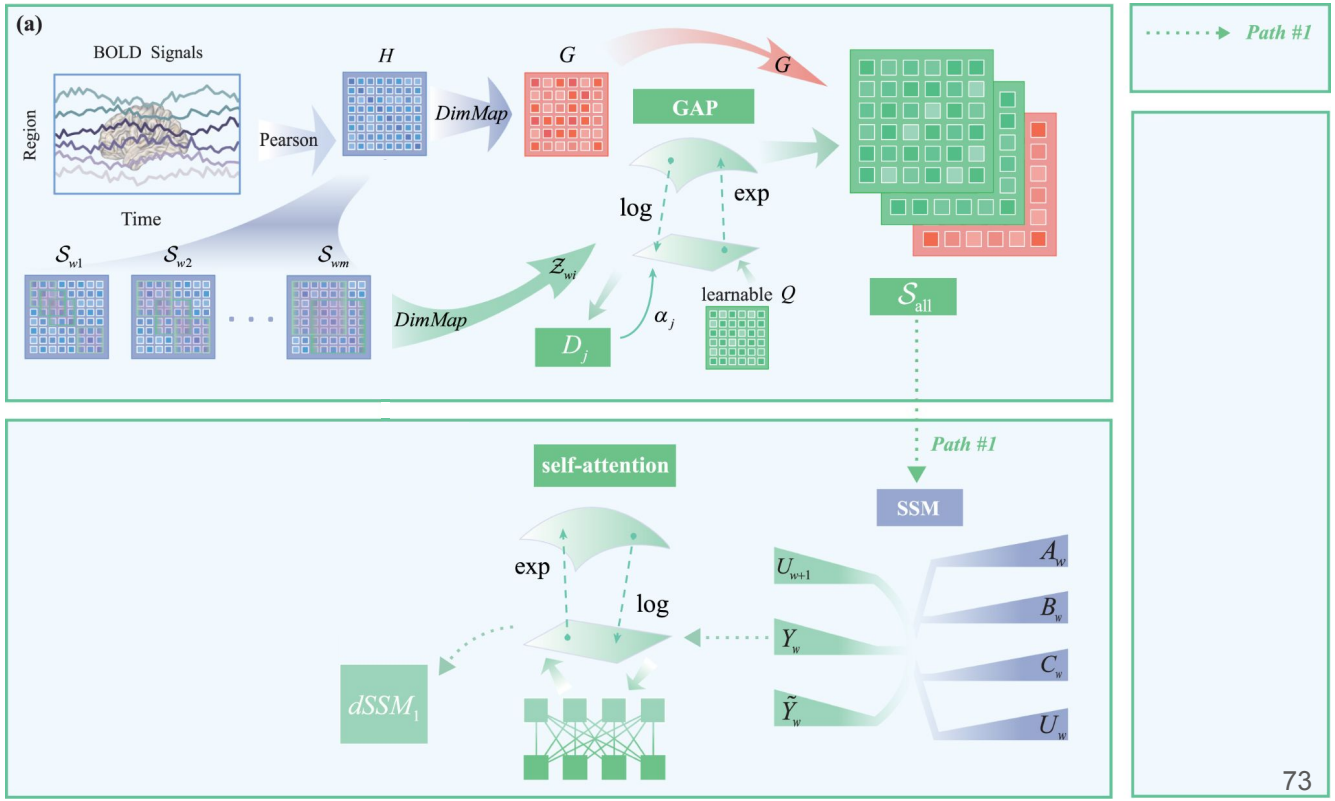
Pseudo-Temporal Sequence Transduction



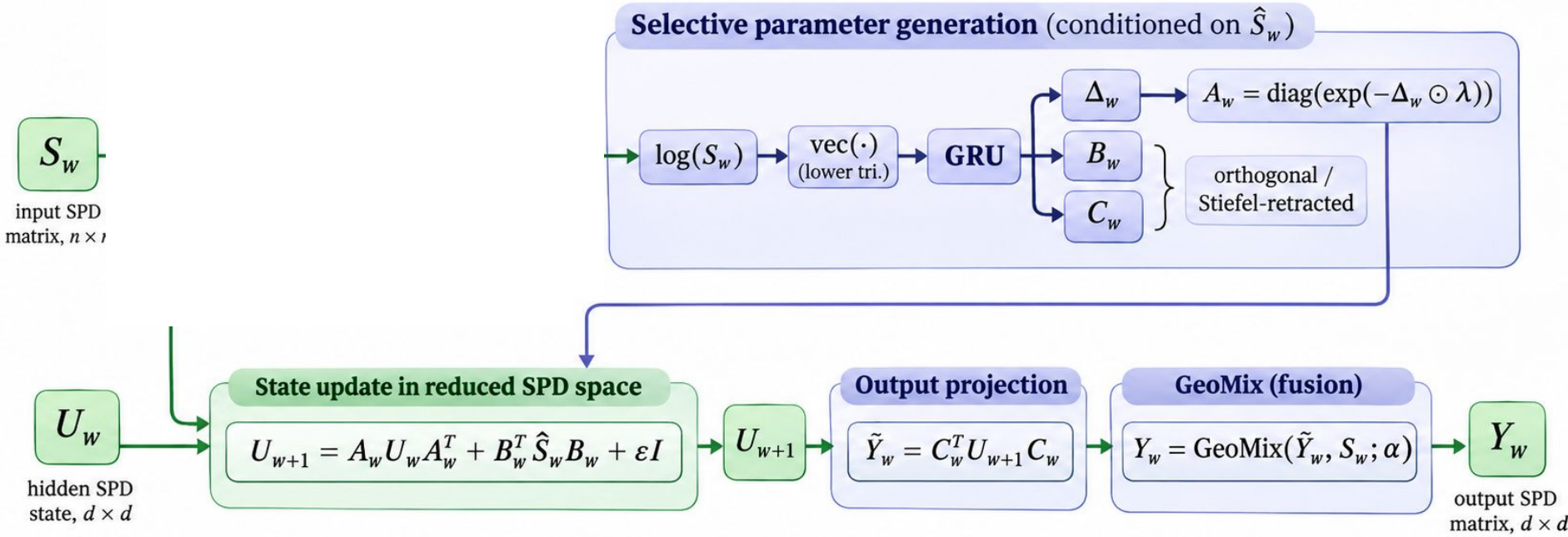
Self-Attention



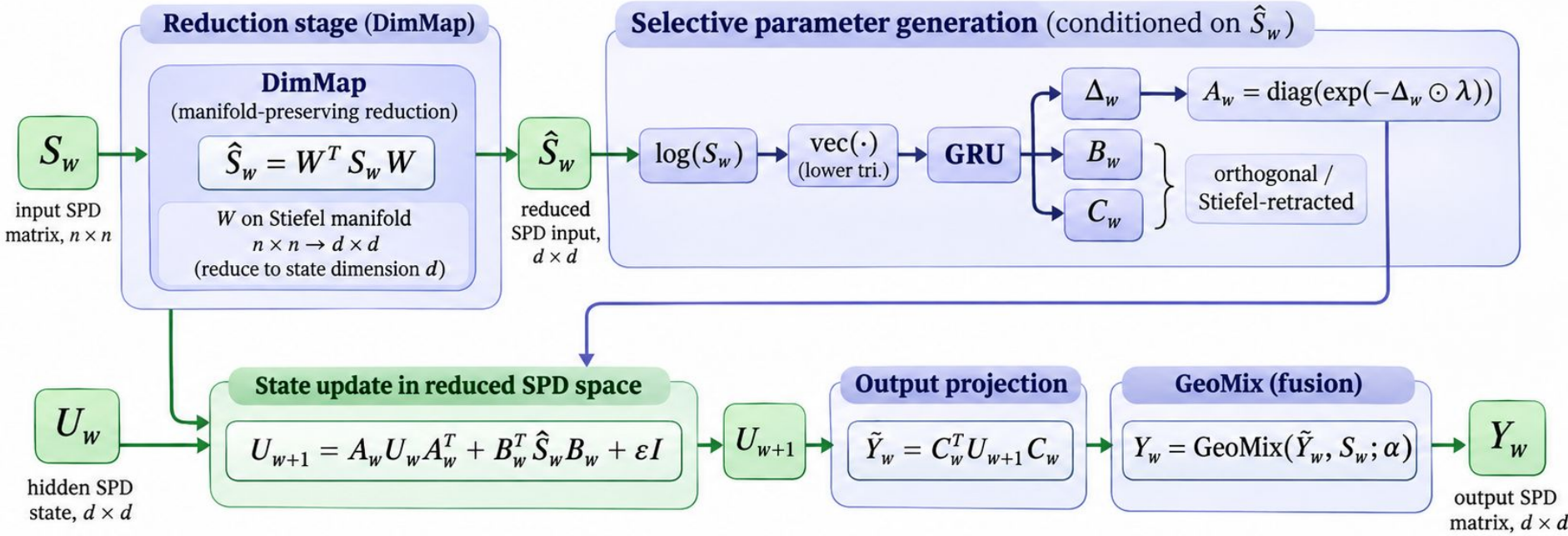
dSSM


(a)
BOLD Signals
Region
Time
Pearson
H
DimMap
G
GAP
G
log
exp
learnable Q
DimMap
D_j
α_j
S_all
Path #1
self-attention
SSM
exp
log
dSSM_1
A_w
B_w
C_w
U_w
Y_w
Ỹ_w
U_w+1
Path #1

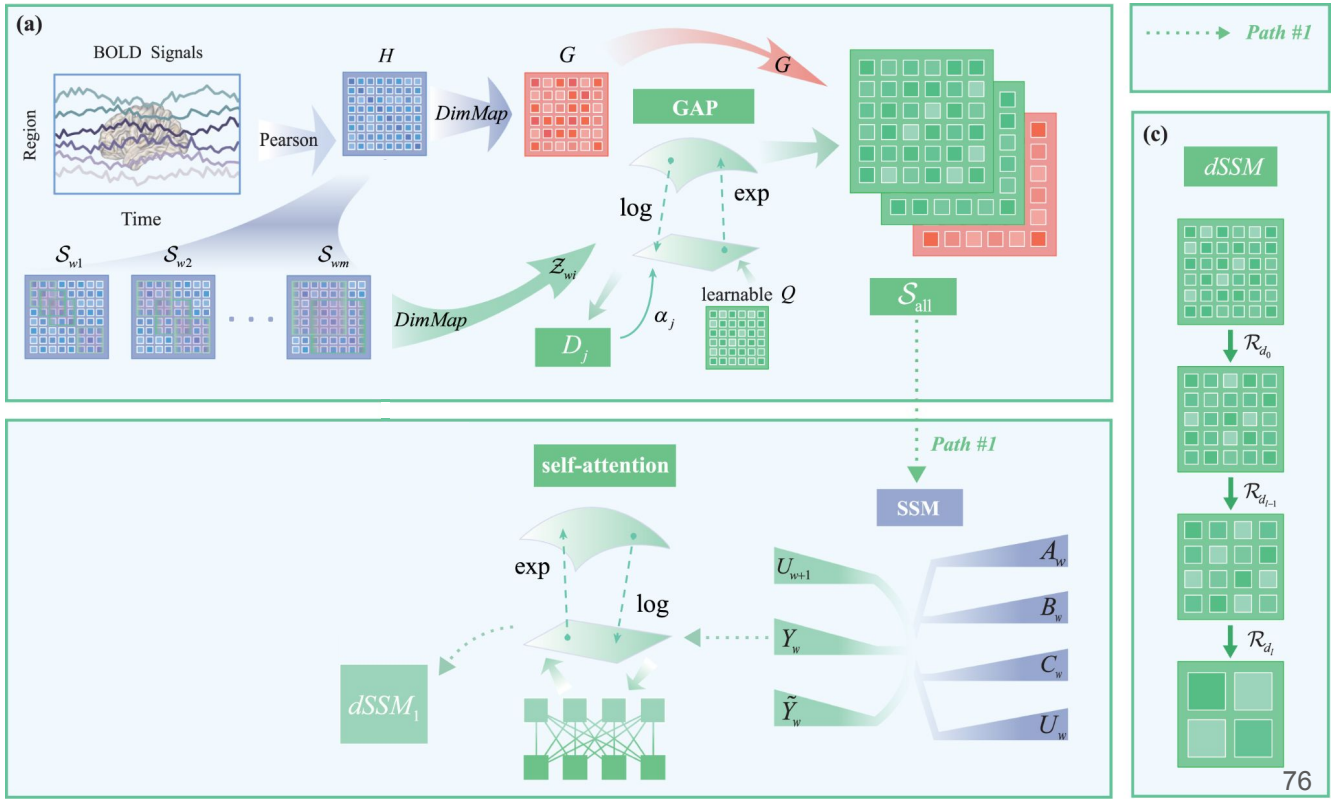
Dimensionality-reducing SSM (dSSM) block on the SPD manifold



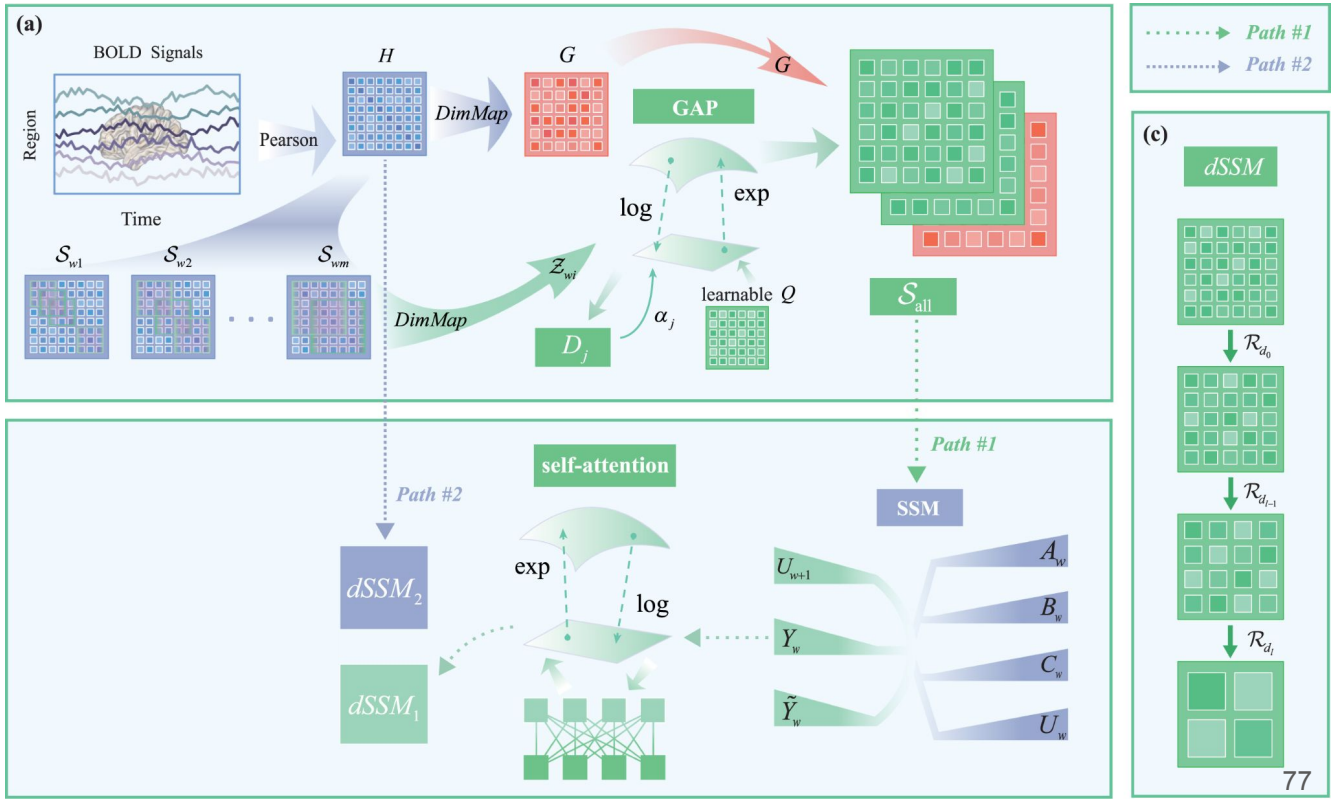
Dimensionality-reducing SSM (dSSM) block on the SPD manifold



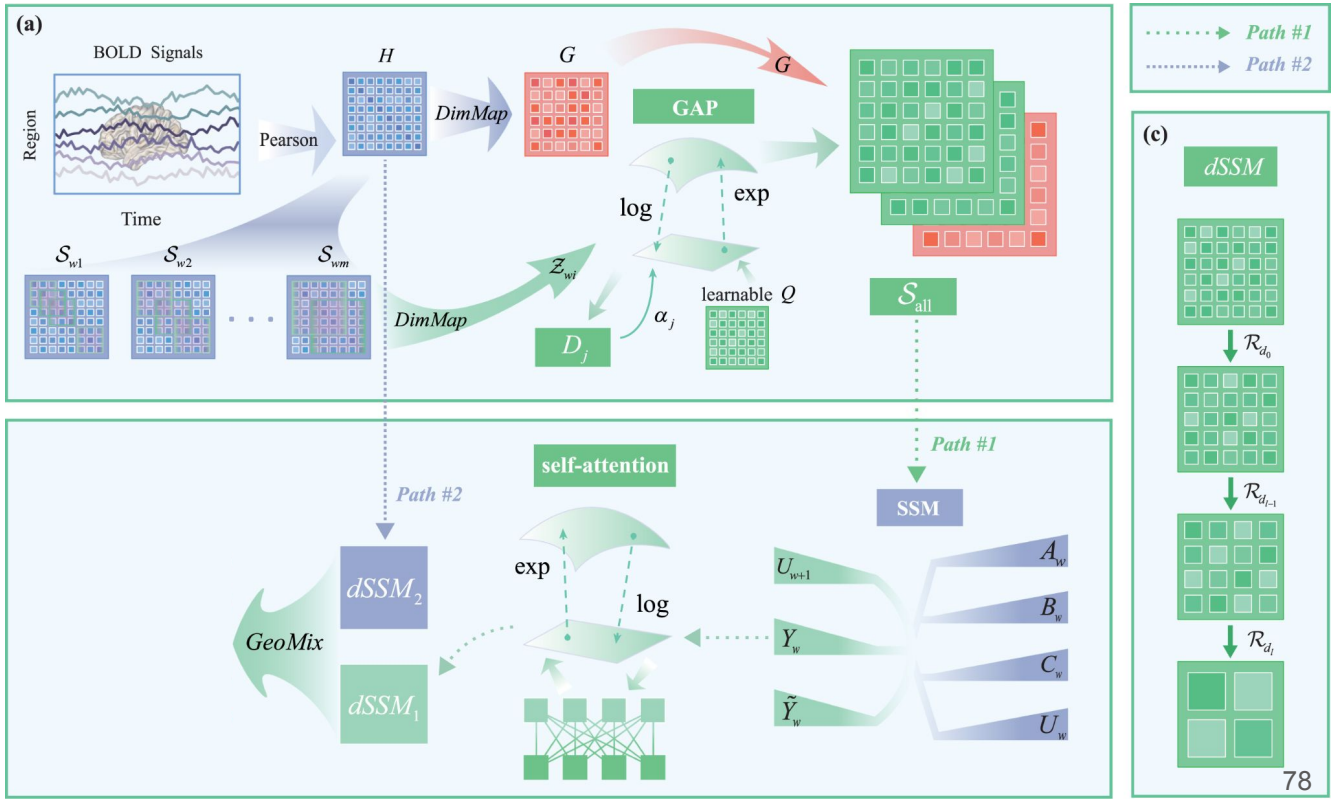
dSSM


(a)
BOLD Signals
Region
Time
Pearson
H
DimMap
G
GAP
G
log
exp
learnable Q
DimMap
D_j
α_j
S_all
Path #1
self-attention
SSM
exp
log
dSSM_1
A_w
B_w
C_w
U_w
Path #1
(c)
dSSM

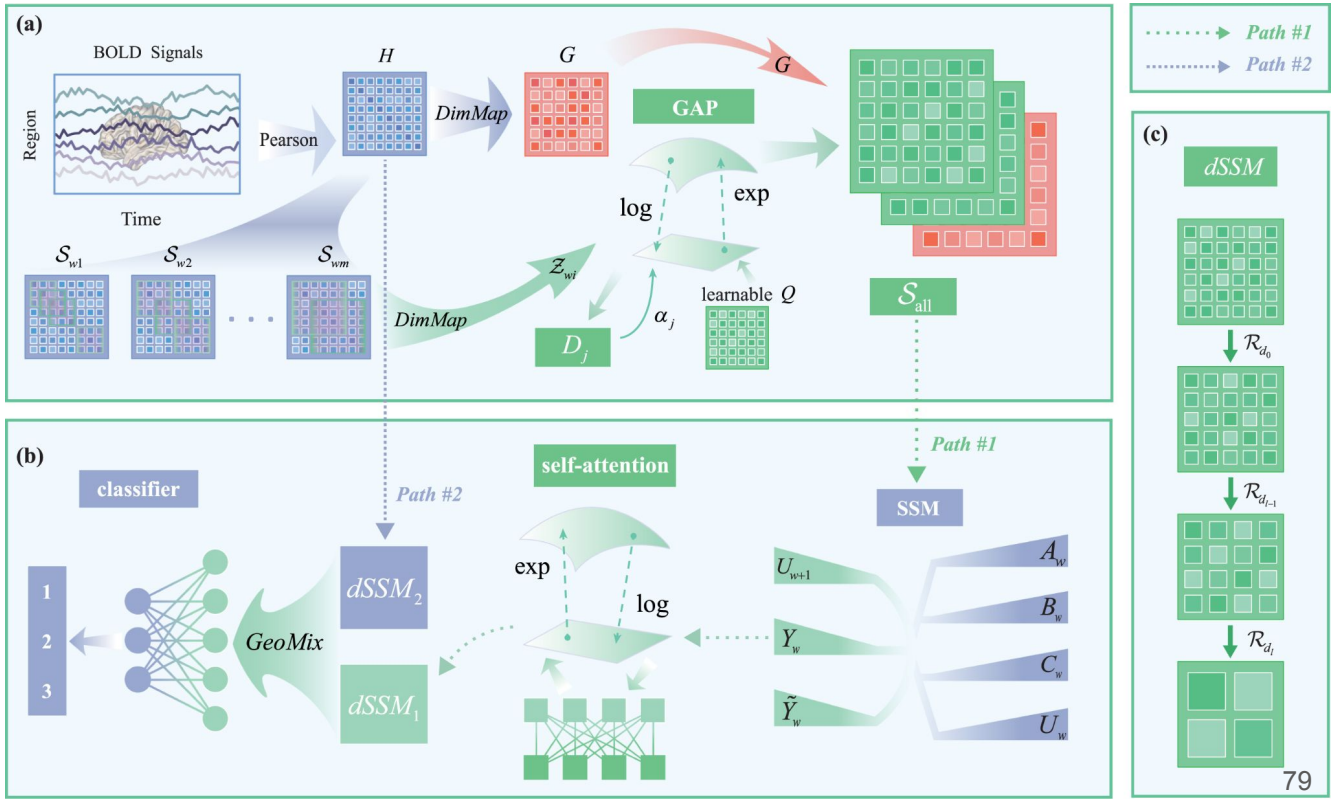
dSSM


(a)
BOLD Signals
Region
Time
Pearson
H
DimMap
G
GAP
G
log
exp
learnable Q
DimMap
α_j
D_j
S_all
Path #1
Path #2
self-attention
SSM
dSSM_2
dSSM_1
exp
log
(c)
dSSM

Path Fusion


(a)
BOLD Signals
Region
Time
Pearson
H
DimMap
G
GAP
G
log
exp
learnable Q
DimMap
α_j
D_j
S_all
Path #1
Path #2
self-attention
SSM
Path #2
exp
log
dSSM_2
dSSM_1
GeoMix
A_w
B_w
C_w
U_w
Path #1
Path #2
(c)
dSSM

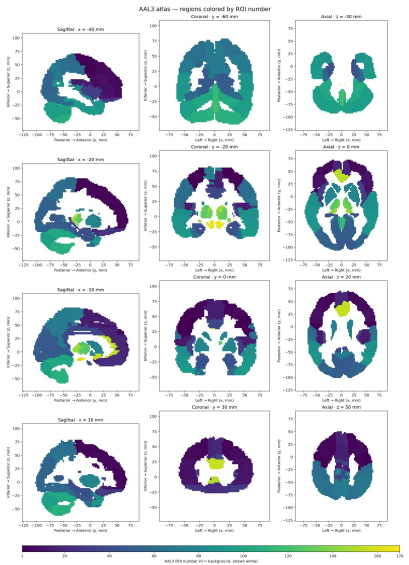
MLP and Output

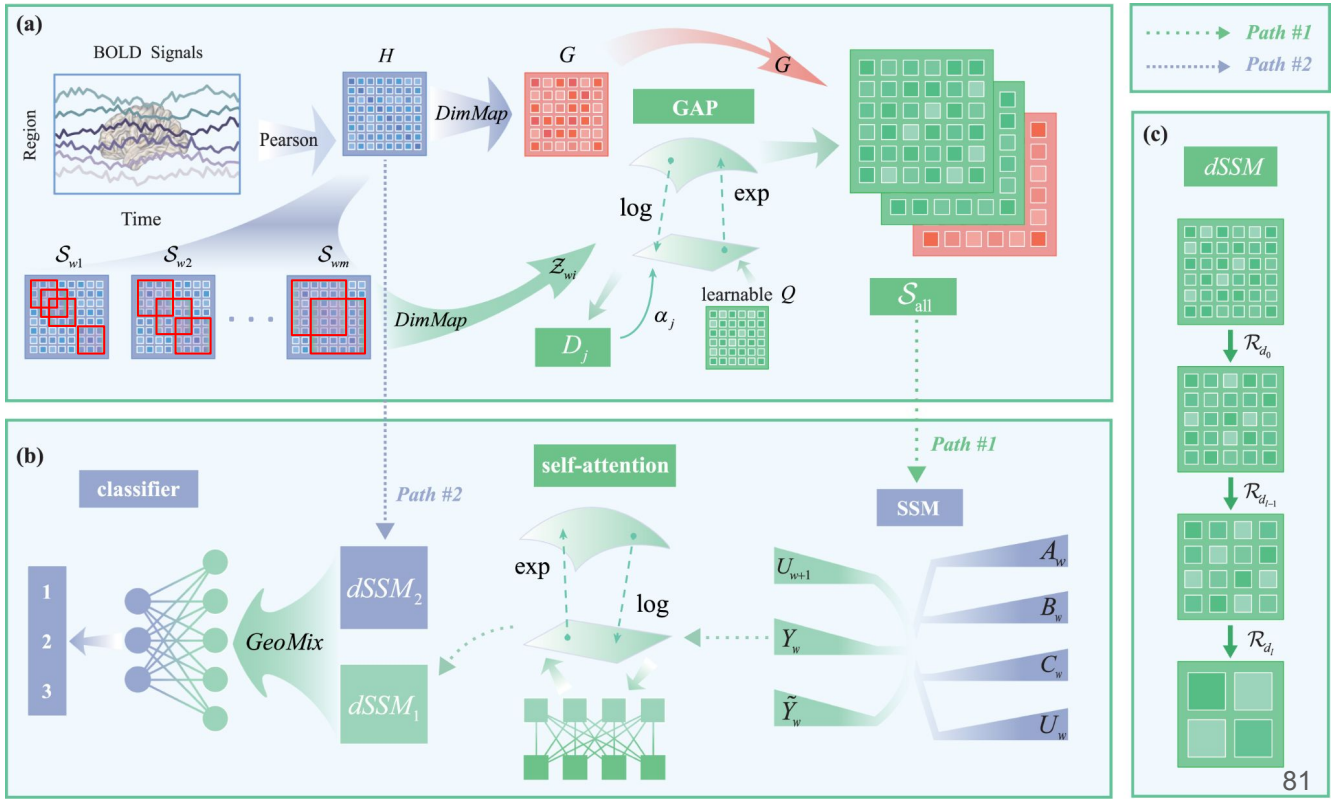


Ablation Studies

Inductive Biases

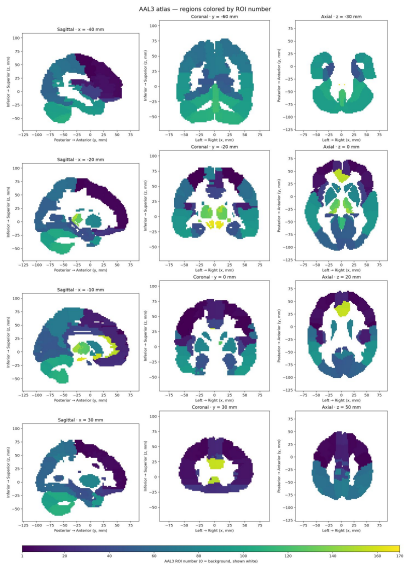
Multiscale connectivity may be useful.

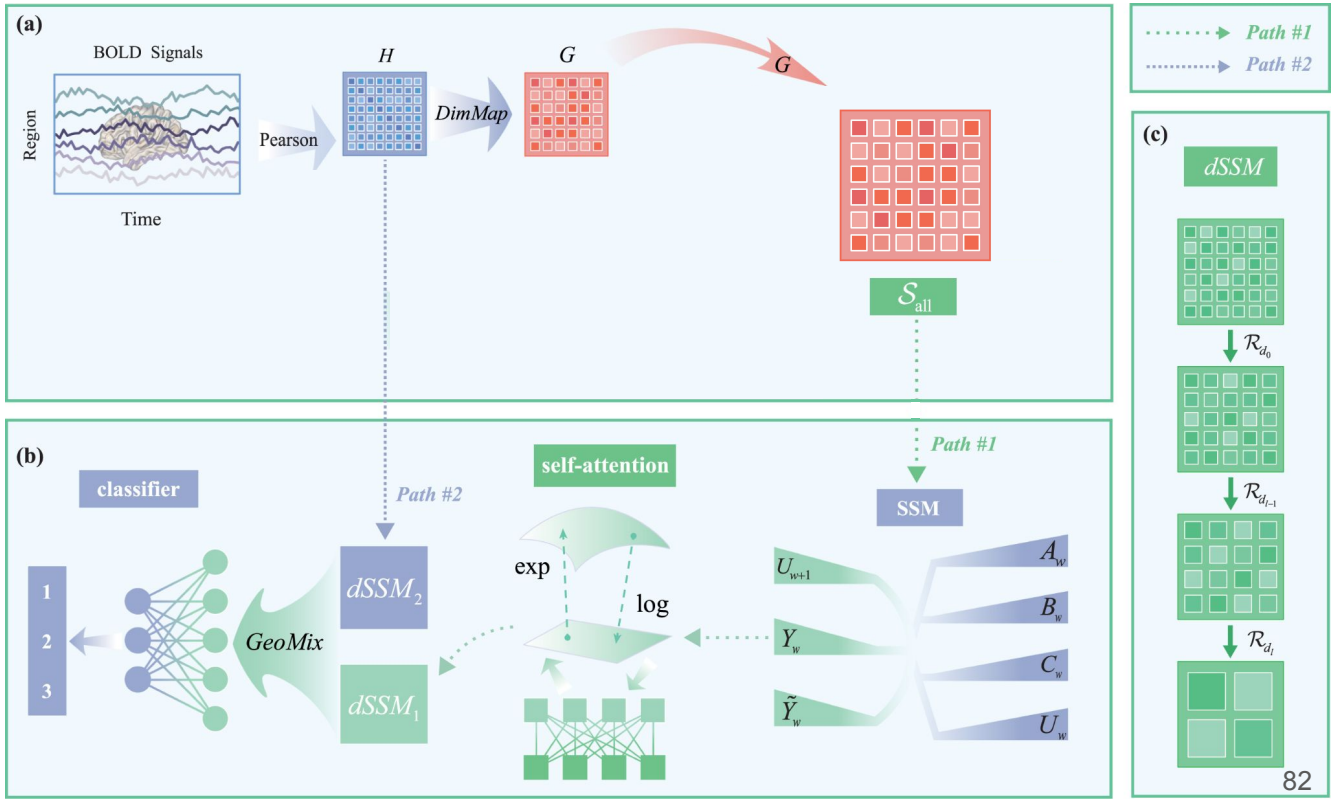




Ablation I: Only G

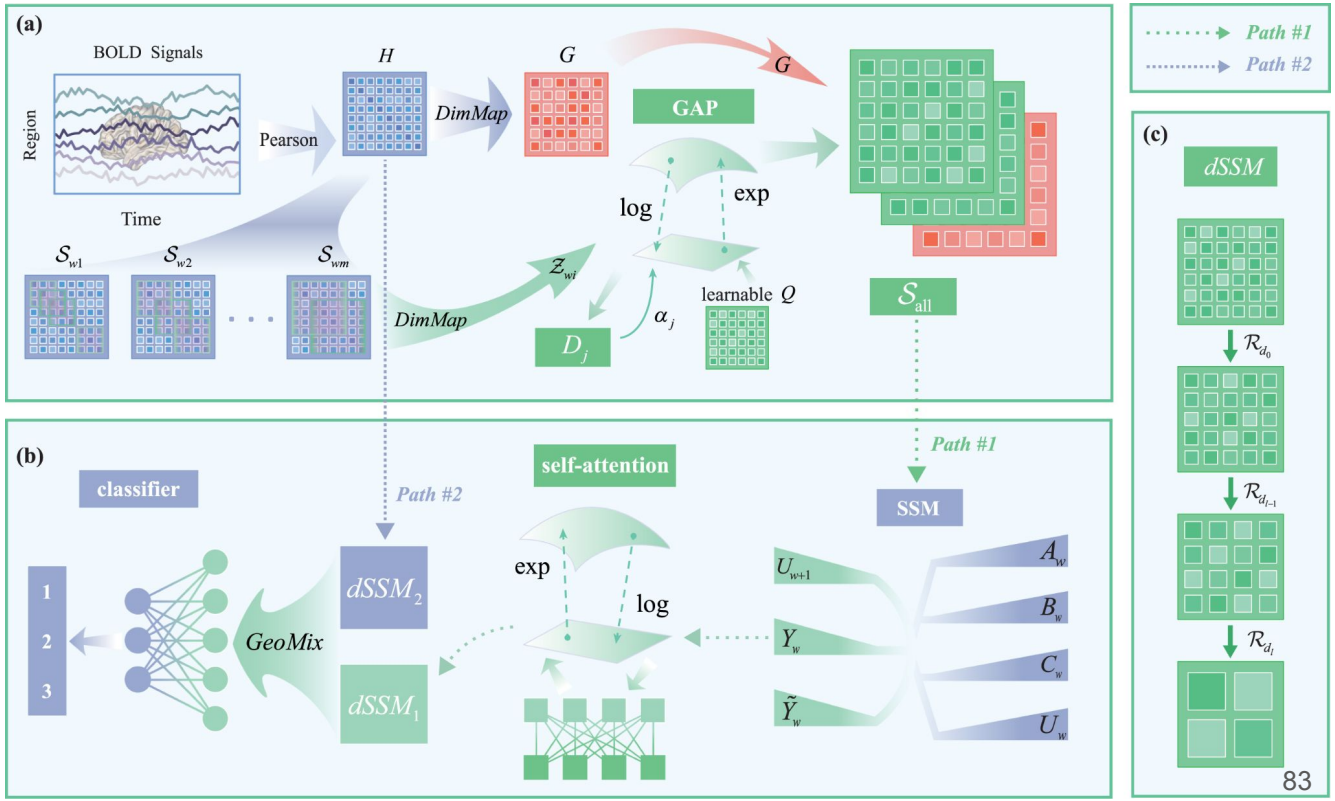
Multiscale connectivity may be useful.





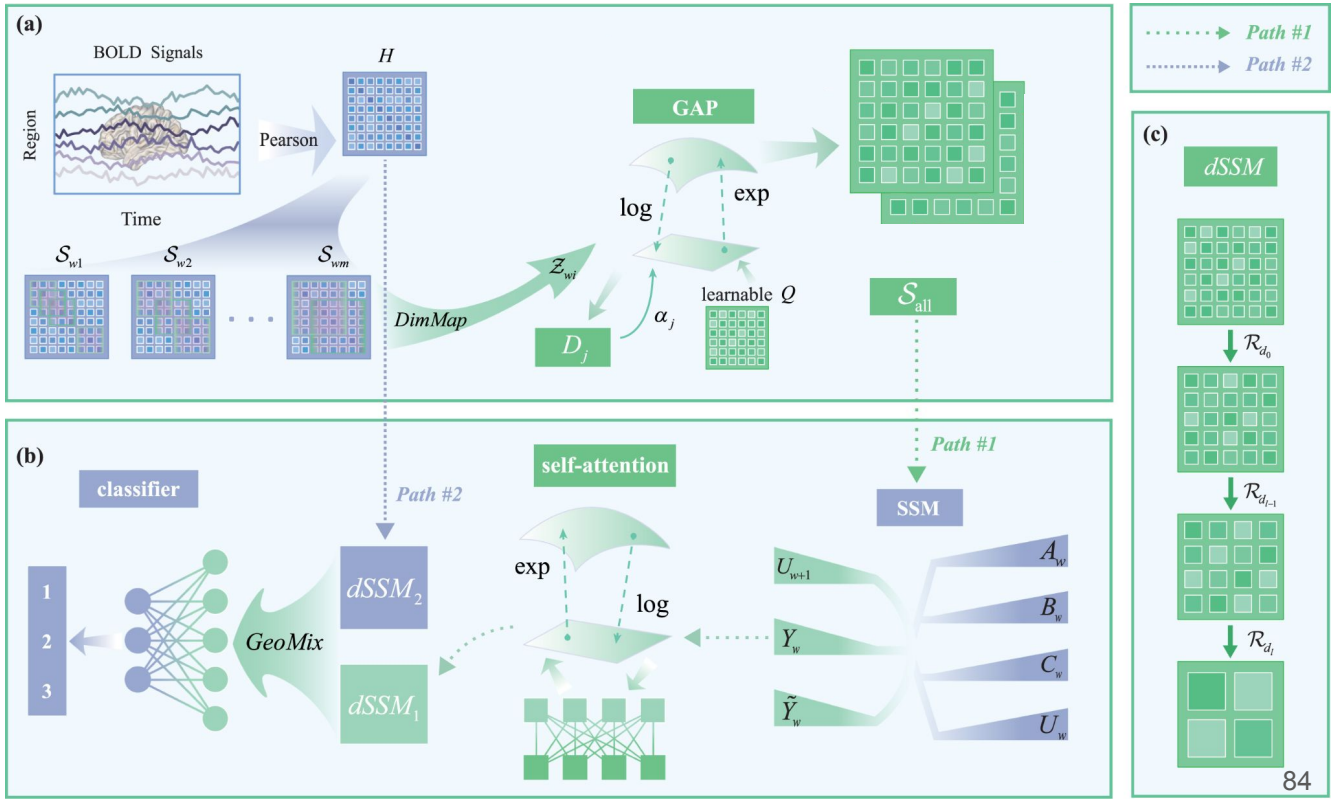
Inductive Biases

Global connectivity provides information beyond local connectivity



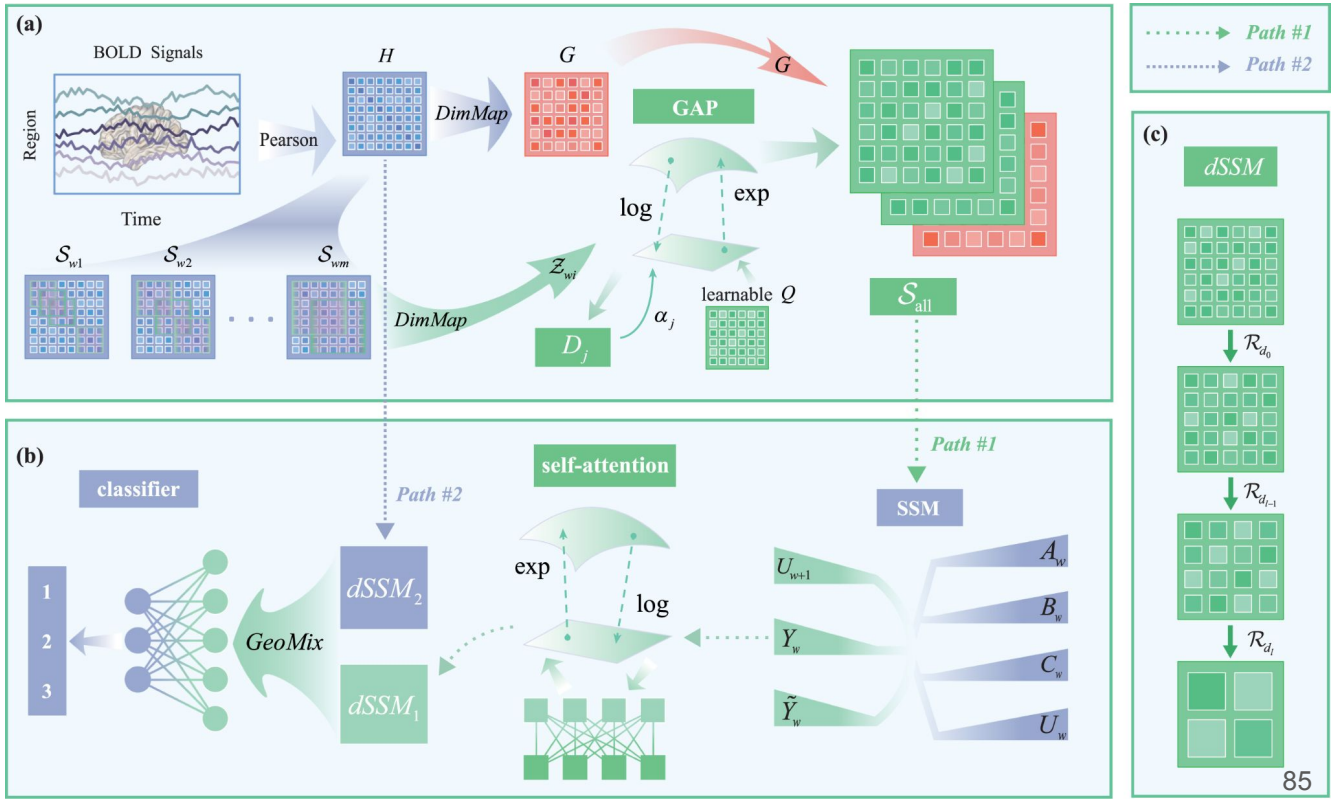
Ablation II: Removing G

Global connectivity provides information beyond local connectivity



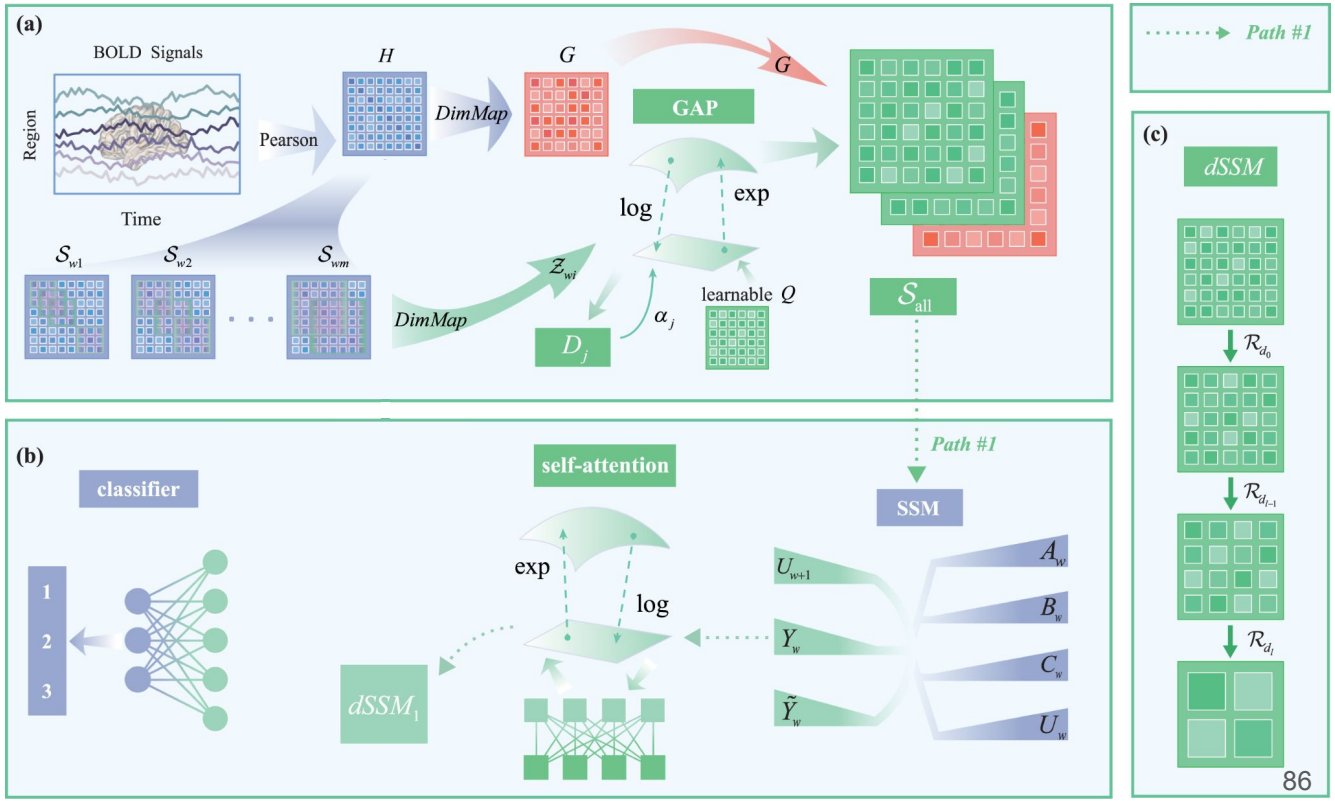
Inductive Biases

Progressive dimensionality reduction is useful.



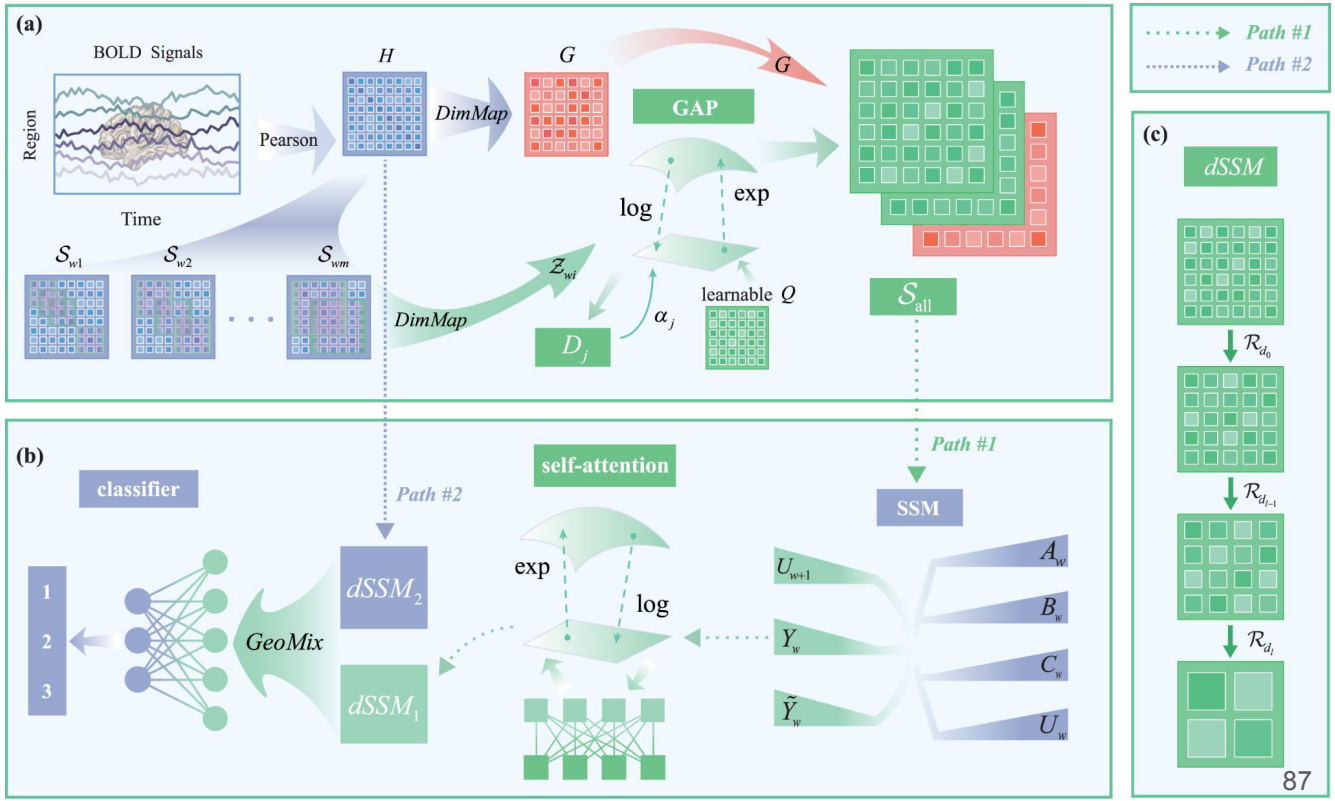
Ablation III: Removing Path #2

Progressive dimensionality reduction is useful.



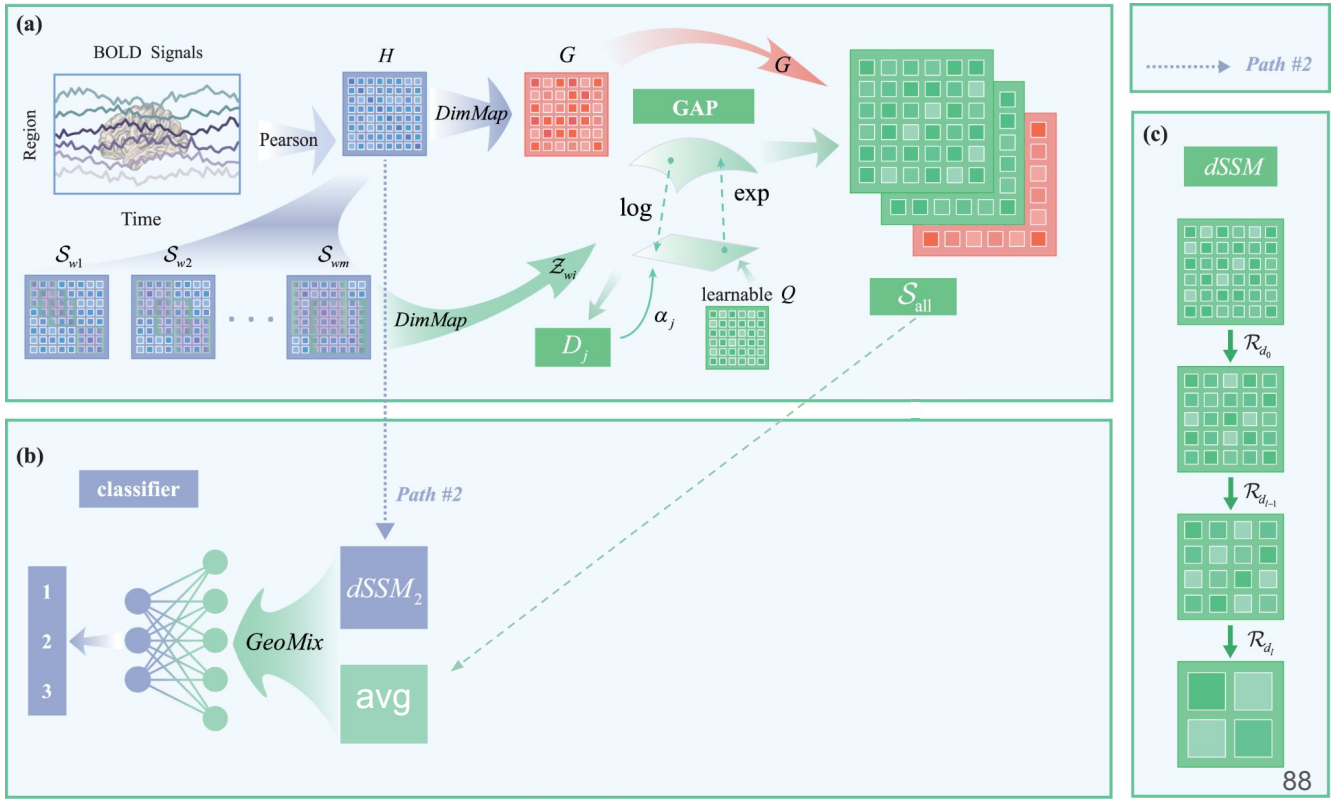
Inductive Biases

The sequence from local to global connectivity is useful.



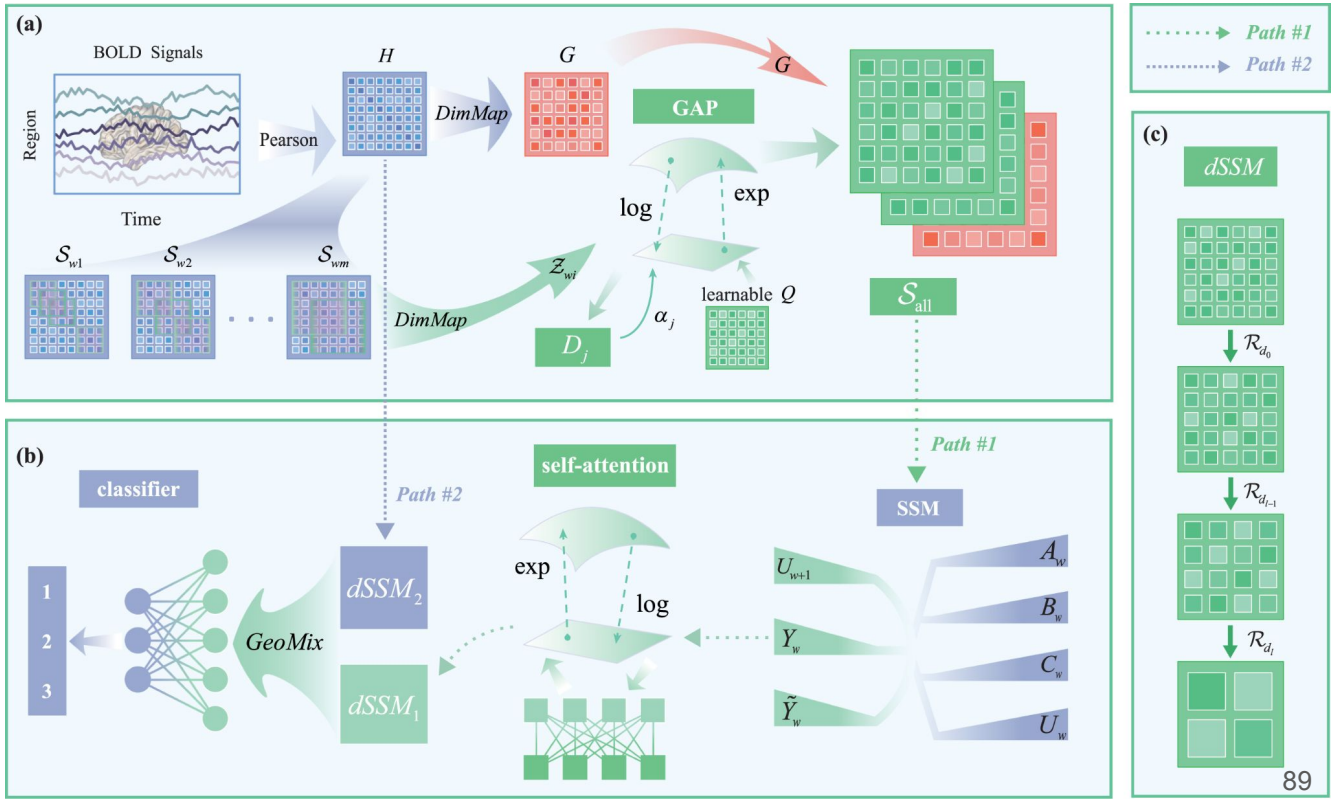
Ablation IV: Removing Path #1

The sequence from local to global connectivity is useful.



Ablation V: Removing Riemannian Mean Normalization

optional stabilization mechanism used in the SSM steps

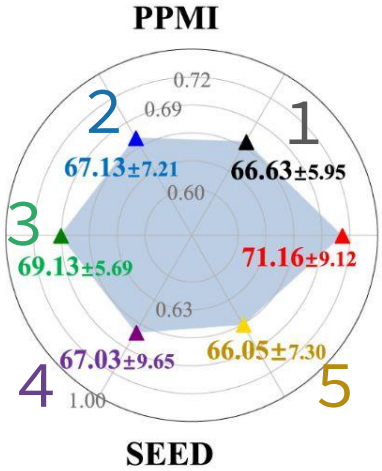


Ablation Results

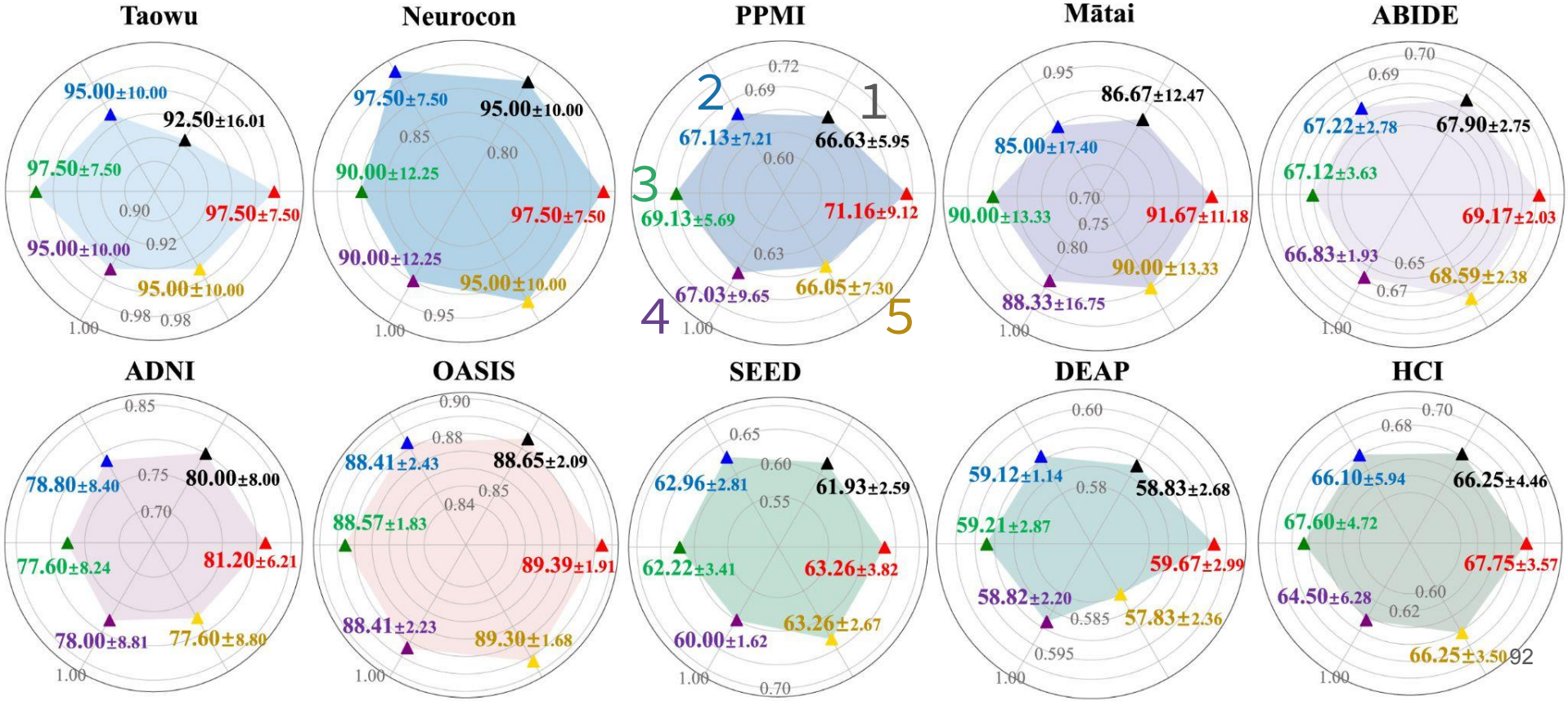
Table B.7
Performance comparison on ablation study across fMRI datasets.

Method	Metric	ADNI	OASIS	PPMI	Taowu	Neurocon	ABIDE	Mātai
Ablation I	Acc	80.00 ± 8.00	88.65 ± 2.09	66.63 ± 5.95	92.50 ± 16.01	95.00 ± 10.00	67.90 ± 2.75	86.67 ± 12.47
	Pre	80.74 ± 8.67	88.81 ± 2.40	65.50 ± 6.48	95.63 ± 13.13	96.67 ± 6.67	68.19 ± 2.90	89.53 ± 9.92
	F1	76.80 ± 9.75	84.97 ± 2.71	60.03 ± 6.16	93.10 ± 13.82	94.67 ± 10.67	67.80 ± 2.81	85.35 ± 13.57
Ablation II	Acc	78.80 ± 8.40	88.41 ± 2.43	67.13 ± 7.21	95.00 ± 10.00	97.50 ± 7.50	67.22 ± 2.78	85.00 ± 17.40
	Pre	81.59 ± 7.78	87.33 ± 1.80	63.36 ± 10.29	91.25 ± 17.50	98.33 ± 5.00	68.11 ± 3.00	82.44 ± 25.18
	F1	75.36 ± 9.38	85.37 ± 3.79	60.63 ± 9.08	92.86 ± 14.29	97.33 ± 8.00	66.94 ± 2.57	81.49 ± 22.78
Ablation III	Acc	77.60 ± 8.24	88.57 ± 1.83	69.13 ± 5.69	97.50 ± 7.50	90.00 ± 12.25	67.12 ± 3.63	90.00 ± 13.33
	Pre	75.36 ± 10.01	87.46 ± 2.72	63.45 ± 9.02	98.75 ± 3.75	93.33 ± 8.16	67.33 ± 3.45	90.36 ± 12.93
	F1	74.24 ± 8.92	86.08 ± 2.14	63.08 ± 6.22	97.67 ± 7.00	89.33 ± 13.06	67.04 ± 3.67	88.78 ± 14.63
Ablation IV	Acc	78.00 ± 8.81	88.41 ± 2.23	67.03 ± 9.65	95.00 ± 10.00	90.00 ± 12.25	66.83 ± 1.93	88.33 ± 16.57
	Pre	75.31 ± 13.75	87.39 ± 2.46	61.08 ± 11.12	91.25 ± 17.50	85.21 ± 19.58	67.52 ± 1.79	90.36 ± 12.93
	F1	72.27 ± 11.61	86.57 ± 2.57	59.94 ± 11.29	92.86 ± 14.29	86.62 ± 16.57	66.22 ± 2.41	87.10 ± 18.10
Ablation V	Acc	77.60 ± 8.80	89.30 ± 1.68	66.05 ± 7.30	95.00 ± 10.00	95.00 ± 10.00	68.59 ± 2.38	90.00 ± 13.33
	Pre	73.11 ± 14.30	88.63 ± 2.05	57.10 ± 13.15	91.25 ± 17.50	96.67 ± 6.67	69.17 ± 2.43	90.36 ± 12.93
	F1	73.17 ± 11.86	87.25 ± 1.12	57.97 ± 9.87	92.86 ± 14.29	94.67 ± 10.67	68.05 ± 2.34	88.78 ± 14.63
Geo-Mamba	Acc	**81.20 ± 6.21**	**89.39 ± 1.91**	**71.16 ± 9.12**	**97.50 ± 7.50**	**97.50 ± 7.50**	**69.17 ± 2.03**	**91.67 ± 11.18**
	Pre	**81.10 ± 8.12**	**88.82 ± 2.32**	**64.81 ± 10.10**	**95.63 ± 13.13**	**98.33 ± 5.00**	**69.29 ± 2.23**	**88.89 ± 17.60**
	F1	**79.08 ± 7.43**	**87.10 ± 2.57**	**65.57 ± 10.17**	**96.43 ± 10.71**	**97.33 ± 8.00**	**68.94 ± 2.03**	**89.48 ± 14.99**

Ablation Results

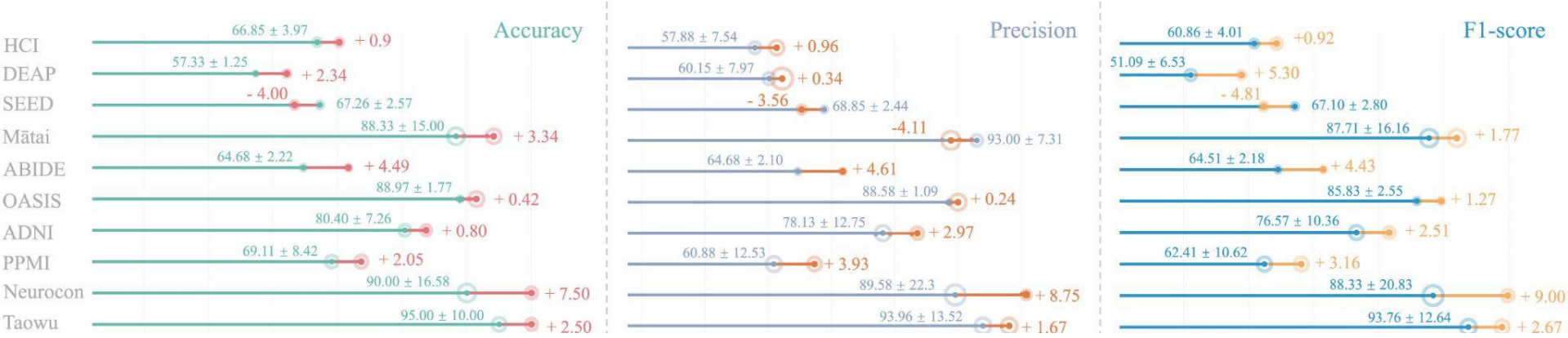


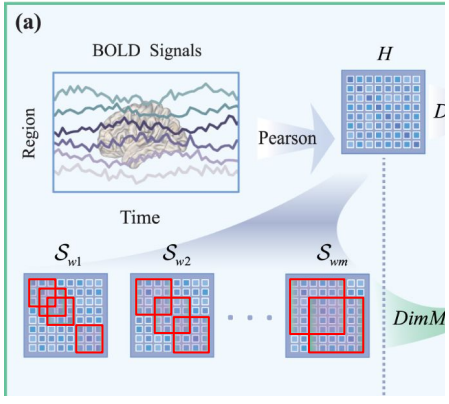
Ablation Results



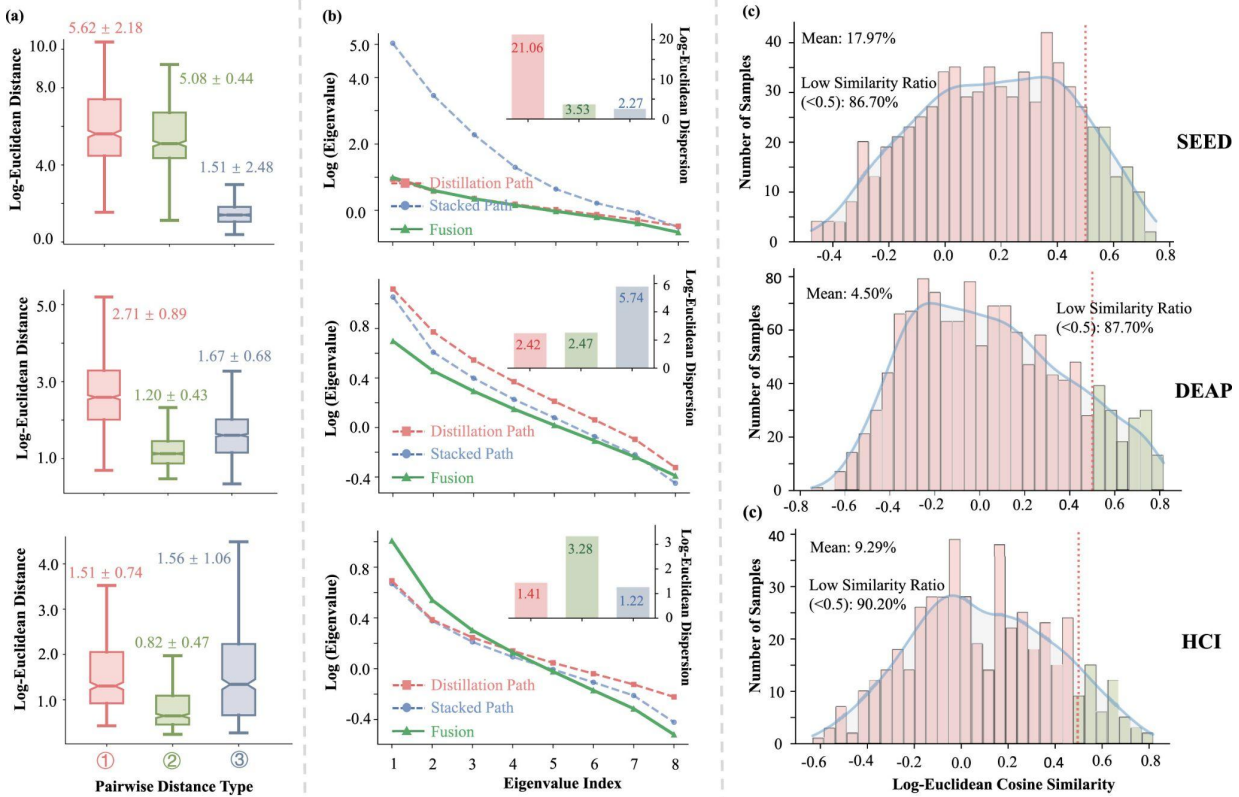
Other Analyses

Pyramid vs Sliding Window

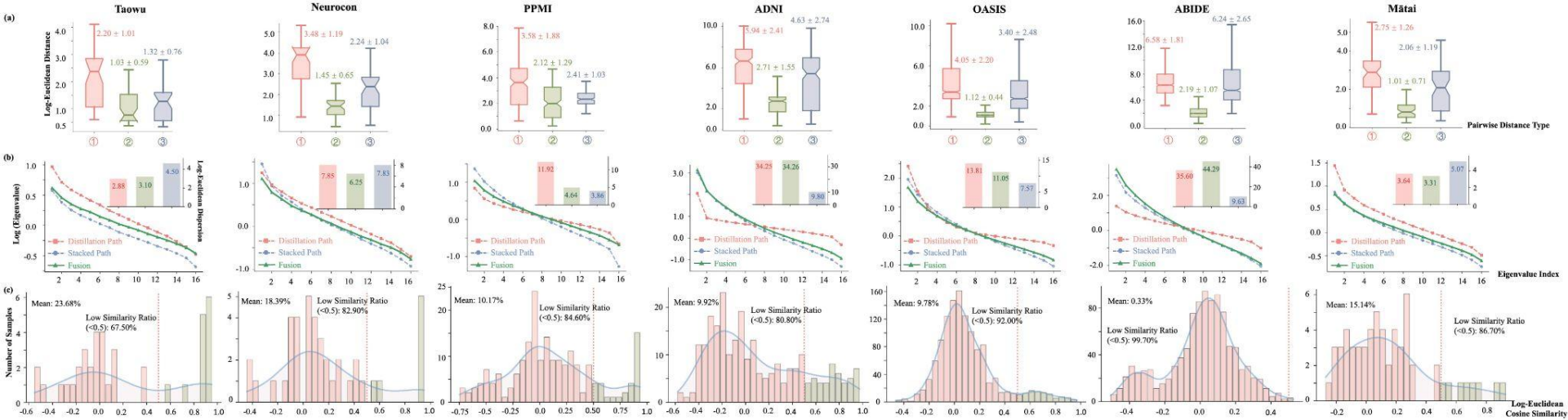




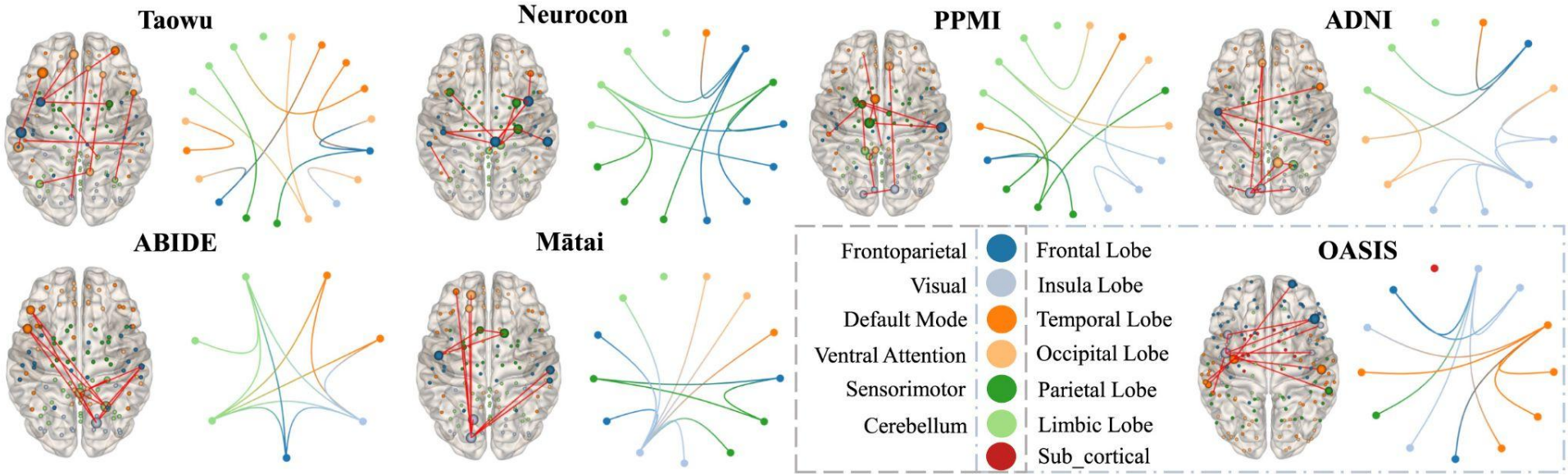
Path 1 vs Path 2



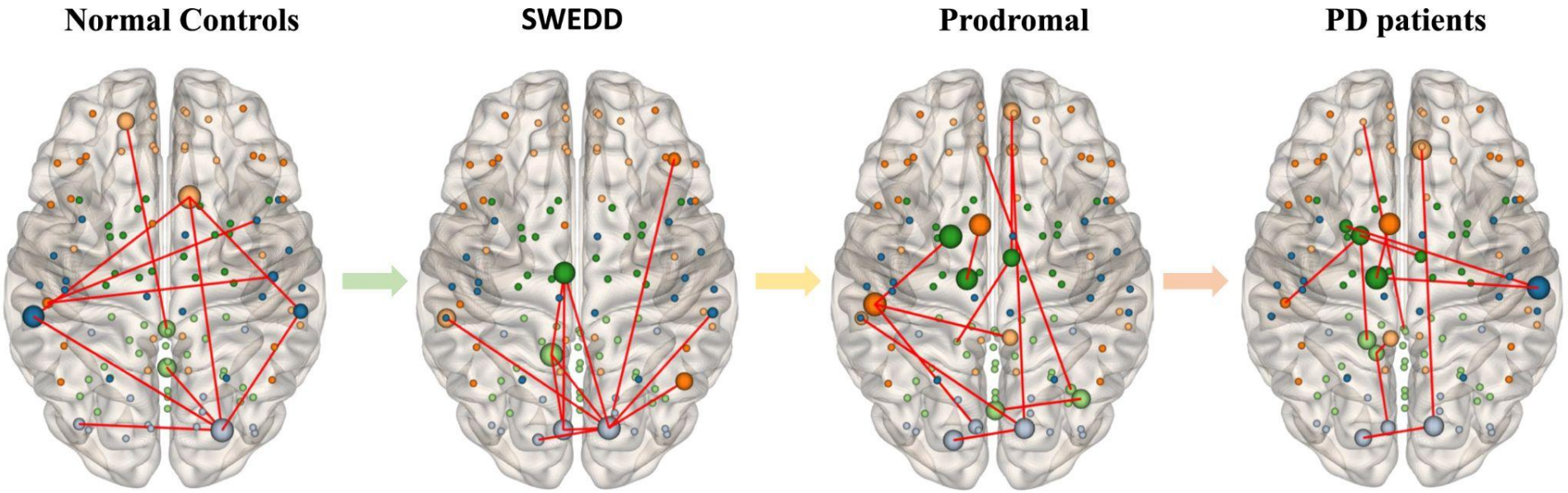
Path 1 vs Path 2



Feature Representation



Feature Representation



Computational Comparison

Table A.6
Computational cost comparisons across different models.

Method	Dataset	Train time /s	Inference time /s	GPU Memory Usage /MB
Transformer	Taowu	0.11	0.01	113.65
	OASIS	25.17	5.87	219.40
	DEAP	23.83	6.41	102.35
Mamba	Taowu	0.18	0.02	52.60
	OASIS	27.46	6.34	178.76
	DEAP	2.31	0.57	18.76
SPDNet	Taowu	0.14	0.02	18.33
	OASIS	26.10	6.25	21.58
	DEAP	2.15	0.37	17.56
Geo-Mamba	Taowu	2.59	0.15	136.07
	OASIS	75.33	13.61	263.27
	DEAP	25.74	2.08	34.55

End